



AG3335 Datasheet

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Feature

Multi –GNSS receiver

- L1 and L5 dual-band GPS receiver with host functionality
- Multi-frequency GPS/GLONASS/Galileo/BeiDou/NavIC receiver
- Support for multi-GNSS including QZSS and SBAS ranging
- Integrated 12 multi-tone active interference cancellers
- RTCM ready (RTCM v2.3 and v3.3)
- Indoor and outdoor multi-path detection and compensation
- RTK, with maximum fixed update rate up to 10Hz

Software features

- EPO™ orbit prediction
- EASY™ self-generated orbit prediction
- LOCUS™ logger function
- Support for time service application, which is achieved by the PPS vs NMEA feature

Microcontroller subsystem

- 530MHz ARM® Cortex®-M4 with FPU and MPU
- MCU operating frequency at 26, 192, 384, and 530MHz
- 17 DMA channels
- One RTC timer, eight 32-bit general purpose timers
- Development support: SWD, JTAG
- Crypto engine
 - AES 128/192/256 bits
 - DES, 3DES
 - MD5, SHA-1/224/256
- True random number generator
- Secure boot, anti-clone, and anti-rollback
- JTAG password protection

Memory

- Up to 664KB SYSRAM, with zero-wait state and 265MHz maximum frequency
- 80kB retention SRAM
- Up to 32KB L1 cache, with high hit rate, zero-wait state and 530MHz maximum frequency
- Embedded 32Mbits flash memory
 - Sleep current 200nA

- Maximum frequency 104MHz
- Embedded 32Mbits pseudo SRAM
 - Sleep current 10μA
 - Maximum frequency 192MHz
- Support for external SPI SRAM
 - Maximum speed 26Mhz
- AESOTF (AES on-the-fly) to secure the data stored in flash

Communication interfaces

- USB 2.0 device
- One SDIO 2.0 master
- Two I2C master (3.4Mbps) and one I2C slave interfaces
- Three UART interfaces (3Mbps, with hardware flow control)
- Two SPI masters and one SPI slave
- Three PWM channels
- 4 channel 10-bit AUXADC (PinMux with GPIO), maximum input voltage 1.8V
- Embedded thermal sensor

Power management

- Integrated LDO regulators with low quiescent current for RTC, RF frontend and GPIOs
- Support for external wakeup
- Over-current and thermal overload protection
- Under voltage lockout protection
- Operating temperature from -40°C to 85°C

Clock source

- Support 26/48MHz TCXO
- Support 32kHz crystal oscillator or internal 32kHz RC for RTC

Package

- 4.2mm x 4.2mm x 1.066mm TFBGA with 0.4mm ball pitch

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1 System Overview

AG3335 series is a highly integrated System-in-Package (SiP), low-power, RFSoc GPS receiver with an application processor, GPS L1 and L5 receiver, a power management unit (PMU), a 4MB flash and a 4MB Pseudo SRAM (PSRAM).

AG3335 series contains an ARM® Cortex®-M4F application processor that can operate in a range of frequencies between 26MHz and 530MHz, and achieve high performance and power efficiency with Level one cache (L1 cache).

The GPS subsystem contains the RF and baseband circuits, which are able to track L1+L5 satellites at the same time, and search GPS satellites using the L1 circuit.

The PMU contains five ultra-low quiescent current low-dropout linear regulators (LDO) to provide a stable power source for the internal and external devices.

The SiP package technology combine the main die, PMU, flash, and PSRAM. This significantly reduces the product size and provides more features in the same space.

1.1 AG3335A product family

The AG3335 product family provides different packages for different market purposes. Table 1.1-1 shows the comparison of the AG3335 series chipsets.

Table 1.1-1. AG3335 product family comparison

Item	AG3335A
Package type	TFBGA (Thin Fine-Pitch Ball Grid Array)
Package size	4.2 x 4.2 x 1.066 mm
Ball count	83-ball
Ball pitch	0.4 mm
GPIO number	37
Power Management Unit (PMU)	Yes
SiP flash (code + data)	Yes
SiP flash size	32Mbits
SiP PSRAM	Yes
SiP PSRAM size	32Mbits
External flash interface (code + data)	–
32kHz clock source	Internal (32K-less) or external
SPI master controller	2 masters
SPI slave controller	1
I2C master	2
I2C slave	1
SDIO 2.0 master	1

Item	AG3335A
PWM number	3
AUXADC	4 channels
Thermal sensor	1 channel

1.2 System architecture

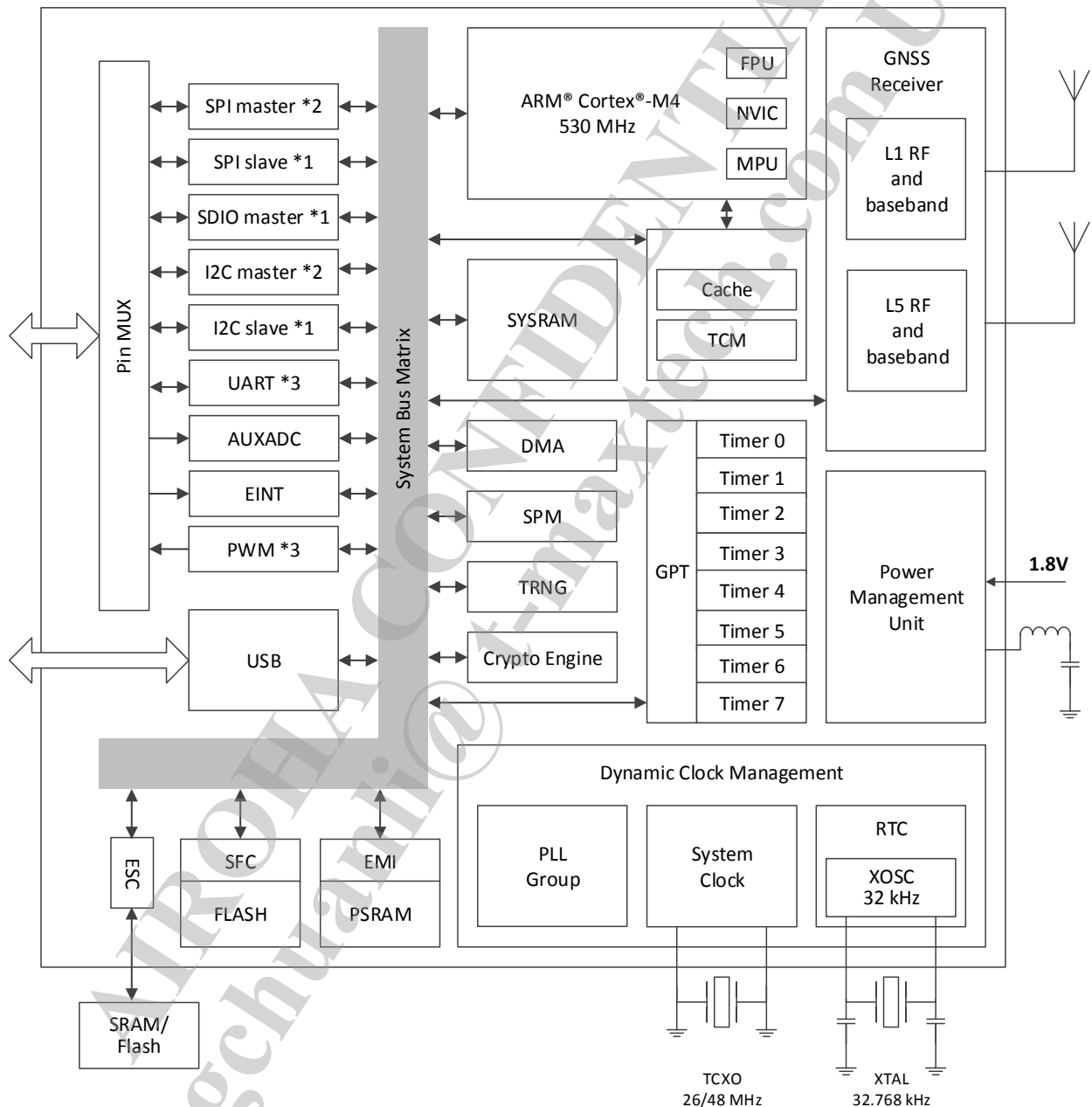


Figure 1.2-1. AG3335A chip architecture

1.3 Platform features overview

1.3.1 Host processor subsystem

- ARM® Cortex®-M4 with Floating Point Unit (FPU) and Memory Protection Unit (MPU)
- MCU operating at 192, 384, and 530Mhz
- Nested Vectored Interrupt Controller (NVIC) with 36 interrupt sources; each with up to 16 levels of priority
 - 192kB memory with zero wait state; It can be programmed as L1 cache or Tightly Couple Memory (TCM)
- Configurable L1 cache size: 32kB, 16kB, 8kB, or 0kB
- SWD and JTAG development support
- Secure boot, anti-clone, and anti-rollback
- JTAG password protection
- True random number generator
- Execute in Place (XIP) on flash memory

1.3.2 Platform

- Dynamic Voltage Scaling (DVS) from SLEEP-Mode to active
- Dynamic Frequency Scaling (DFS)
- 17 DMA channels
- RTC timer
- 8 general purpose timers (GPTs)
- Watchdog timer (WDT)
- Crypto engine for AES/SHA1/SHA224/SHA256
- True random number generator

1.3.3 Memory

- Low latency 664kB system RAM (SYSRAM) with a maximum speed of 265MHz
- 80kB retention SRAM
- System in Package (SiP) 32Mb low-power flash
 - Maximum speed 104MHz
 - 200nA sleep current
- System in Package (SiP) 32Mb low-power Pseudo SRAM (PSRAM) (only in AG3335A)
 - Maximum speed 192Mhz
 - 10µA sleep current
 - AESOTF (AES on-the-fly) for secure data storage
- Support external SPI SRAM
 - Maximum speed 26Mhz
 - AESOTF (AES on-the-fly) for secure data storage

2 Functional Description

2.1 ARM Cortex®-M4 host processor subsystem

2.1.1 ARM® Cortex®-M4 with FPU

The Cortex-M4 with FPU is a low-power processor with 3-stage pipeline Harvard architecture. It has reduced pin count and low power consumption and delivers very high-performance efficiency and low interrupt latency, making it ideal for embedded microcontroller products.

The processor incorporates:

- IEEE754-compliant single-precision floating-point computation unit (FPU).
- A Nested Vectored Interrupt Controller (NVIC) to achieve low latency interrupt processing.
- Enhanced system debugging with extensive breakpoint.
- An optional Memory Protection Unit (MPU) to ensure platform security robustness.

The Cortex-M4 executes the Thumb®-2 instruction set with 32-bit architecture, with the high code density of 8-bit and 16-bit microcontrollers. The instruction set is fully backward compatible with Cortex-M3/M0+.

AG3335 series has further enhanced the Cortex-M4 with FPU to reduce the power by another 11% (in Dhrystone) compared to the original Cortex-M4. Low power consumption is a significant feature for IoT and Wearables application development.

2.1.2 Cache controller

A configurable 32kB cache is implemented to improve the code fetch performance when CPU accesses a non-zero wait-state memory such as EMI, external flash or boot ROM through the on-chip bus.

The core cache is a small block of memory containing a copy of a small portion of cacheable data in the external memory. If CPU reads a cacheable datum, the datum will be copied to the core cache. When CPU requests the same datum again, it can be obtained directly from the core cache (called cache hit) instead of fetching it again from the external memory to achieve zero wait-state latency.

The cache can be disabled and this block of memory can be turned into tightly coupled memory (TCM), a high-speed memory for normal data storage. The sizes of TCM and cache can be set to one of the following four configurations:

- 32kB cache, 160kB TCM
- 16kB cache, 176kB TCM
- 8kB cache, 184kB TCM
- 0kB cache, 192kB TCM

2.1.3 Memory management

Three types of memories are implemented for use:

- On-die memories (SRAMs) with up to 32kB at CPU clock speed with zero wait state.
- Embedded flash of 32Mbits to store programs and data
- Embedded pseudo SRAM (PSRAM) of 32Mbits for application storage

The SRAMs are composed of TCMs and L1 caches. The L1 cache (up to 32kB) is implemented to improve processor access performance of the long latency memories.

TCMs are designed for applications requiring high speed, low latency and low power consumption. Each TCM has its own power state: active; retention; or power-down. TCM must be in the active state for normal read and write access. The retention state saves the SRAM content and consumes the minimum of leakage current with no access. Power-down state loses the content and consumes almost zero power.

The TCMs can also be accessed by other internal AHB masters (e.g., DMA or the multimedia subsystem) for low power applications. These applications can run on TCM without powering on PSRAM or flash to save more power.

Boot ROM is also implemented for processor boot-up and its content is unchangeable.

2.1.4 Memory protection unit (MPU)

The MPU is an optional component to manage the CPU access to memory. The MPU provides full support for:

- Protection regions (up to eight regions, which can be further divided into 8 sub-regions).
- Overlapping protection regions, with region priority.
- Access permissions.
- Exporting memory attributes to the system.

The MPU is useful for applications where a critical code must be protected against the misbehavior of other tasks. It can be used to define access rules, enforce privilege rules, and separate processes.

2.1.5 Nested Vectored Interrupt Controller (NVIC)

The NVIC supports up to 36 maskable interrupts and four interrupt lines of Cortex-M4 with 16 priority levels. The NVIC and the processor core interface are closely coupled to enable low latency interrupt processing and efficient processing of late arriving interrupts. The NVIC maintains knowledge of the stacked or nested interrupts to enable tail-chaining of interrupts. The processor supports both level and pulse interrupts with programmable active-high or low control.

2.1.6 External Interrupt Controller

The external interrupt controller (EINT) consists of up to 32 edge detectors to generate event or interrupt requests. Each input line can be independently configured to select the type (event or interrupt) and the corresponding trigger event (e.g., rising edge, falling edge, both, or level). Each line can also be masked independently. A pending register maintains the status line of the interrupt requests.

2.2 Platform description

2.2.1 Boot mode

While the chip is starting up, the on-chip boot ROM is executed to determine the next booting sequence. The options are flash download, normal boot mode, or external boot mode.

- Flash download mode – The bootloader is part of the flash and can be reprogrammed through the UART, USB, or I2C interface.
- Normal boot up mode – Boot ROM copies the bootloader from the flash to the internal memory without going into flash download mode. When the system completes the boot ROM copy operation, it jumps to the bootloader and executes it.
- External boot up mode – Boot ROM copies the bootloader from the host chip to the internal memory via the SPI interface. When the system completes the boot ROM copy, it jumps to the bootloader and executes it.

2.2.2 Trapping and mode selection

Mode selection	Trapping pin	Description	Trapping condition
XTAL frequency	GPIO12	GND: 26Mhz (default) DVDD18_IO: 48Mhz	Power-on reset
XTAL type	GPIO25	GND: TCXO (default) DVDD18_IO: DCXO (reserved)	Power-on reset
Slow clock source	GPIO4	GND: external 32K clock source is enabled DVDD18_IO: internal (32K clock is divided from TCXO or DCXO)	Power-on reset
Download mode	GPIO21, GPIO20	Decide if firmware needs update {GND,GND}: UART download {GND,DVDD18_IO}: USB download {DVDD18_IO,GND}: SPI download {DVDD18_IO, DVDD18_IO}: I2C download	Power-on reset
XBOOT enable	GPIO11	GND: disable host interface at boot (default) DVDD18_IO: enable host interface via SPI slave at boot	Power-on reset
System Level Test (SLT) mode	GPIO24	Airoha in-house testing purpose GND: Normal mode (default) DVDD18_IO: SLT mode	Power-on reset

Table 2.2-1. Trapping pin list

2.2.2.1 Trapping pin characteristic

Trapping pins	default pull direction	Internal pull resistor value	Recommended external pull circuit for default trapping	Recommended external pull circuit for non-default trapping
GPIO4	Internal pull down	8.33kΩ	N.C. or pull down	Pull up with 1kΩ
GPIO11	Internal pull down	8.33kΩ	N.C. or pull down	Pull up with 1kΩ
GPIO12	Internal pull down	75kΩ	N.C. or pull down	Pull up with 10kΩ
GPIO20	Internal pull down	75kΩ	N.C. or pull down	Pull up with 10kΩ
GPIO21	Internal pull down	75kΩ	N.C. or pull down	Pull up with 10kΩ
GPIO25	Internal pull down	75kΩ	N.C. or pull down	Pull up with 10kΩ

Table 2.2-2. Trap pin electric characteristics

2.2.2.2 Trapping timing diagram:

The GPIO value is latched into trapping mode when the power-on reset de-assert. The process shown in the following image goes from low to high.

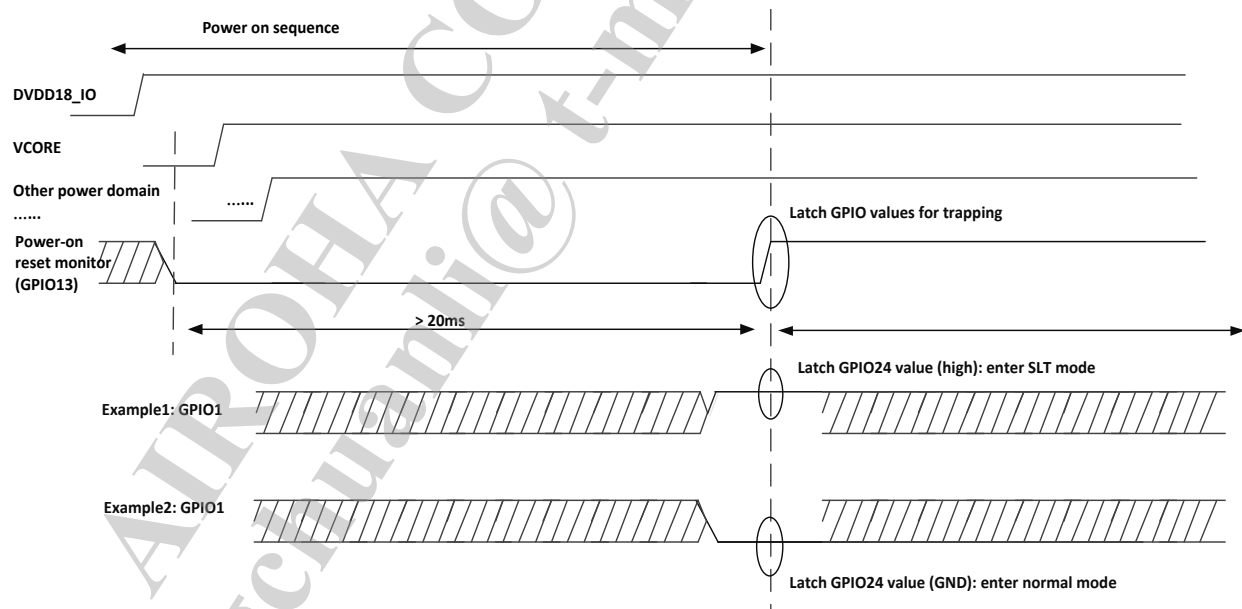


Figure 2.2-1. Trapping timing diagram (SLT mode as example)

The SLT mode is for Airoha in-house testing purposes. Please make sure not to go into SLT mode or the boot up system will fail.

The power-on reset can be observed on the pin GPIO13 (RESETB_OUT) in AG3335.

2.2.3 Bus architecture

In AG3335, the main system consists of a 32-bit multilayer AHB bus matrix that interconnects:

- Masters:
 - Cortex-M4 with floating point unit (FPU) core I-bus, D-bus and S-bus
 - L1 GPS
 - L5 GPS
 - DMA memory bus, including I2C SLV
 - Crypto Engine DMA bus
 - Serial Peripheral Interface DMA bus (SPI)
 - USB DMA bus
 - Secure Digital Input/ Output DMA bus (SDIO)
- Slaves:
 - Pseudo SRAM controller (PSRAM)
 - Embedded flash
 - External SRAM controller (ESC)
 - Main System SRAM (SYSRAM)
 - Retention RAM
 - Tightly-coupled memory (TCM)
 - AHB peripheral including AHB to APB bridges and APB peripherals
 - L1 GPS slaves
 - L5 GPS slaves

The bus matrix provides access from a master to a slave, enabling concurrent access and efficient operation even when several high-speed peripherals are working simultaneously. The AG3335 bus connectivity is shown in Table 2.2-3.

Table 2.2-3. AG3335 bus connection

Slave Master	EMI (PSRAM)	Flash	ESC	SYSRAM	Retention RAM	TCM	AO_APB	PD_APB	L1 GPS	L5 GPS
CM4	•	•	•	•	•	•	•	•	•	•
L1 GPS	•	•	•	•	•	•	•	•	•	•
L5 GPS	•	•	•	•	•	•	•	•	•	•
MCU DMA	•	•	•	•	•	•	•	•	•	•
I2C SLV DMA	•	•	•	•	•	•	•	•	•	•
Crypto Engine	•	•	•	•	•	•	•	•	•	•
USB	•	•	•	•	•	•				
SDIO	•	•	•	•	•	•				
SPI MST	•	•	•	•	•	•				
SPI SLV				•						

2.3 Clock source and structure

2.3.1 Clock architecture

The clock controller (see Figure 2.3-1) distributes the clocks from different oscillators to the core circuit and the peripherals. It also manages clock gating for low-power modes and ensures clock robustness. The clock controller features:

- **Clock prescaler** — provides the best trade-off between speed and current consumption. A programmable prescaler can adjust the clock frequency to the CPU and peripherals.
- **Safe clock switching** — clock sources can be adjusted during runtime through a configuration register.
- **Clock management** — the clock controller can stop the clock of the core circuit, individual peripherals or memory to reduce the power consumption. The AHB and APB clocks support dynamic clock slow down or gating when bus fabric is idle.
- **System clock source** — three different clock sources can be used to drive the system clock (F_{SYS}):
 - 26MHz Temperature Compensate Crystal Oscillator (TCXO) that can supply reference clock for PLLs.
 - UPLL fed by reference clock TCXO, with a maximum frequency at 312MHz.
 - Low-power RC oscillator (LPOSC) — low-power consumption with large frequency variation.
- **Auxiliary clock source** — there are three low-power clock sources to drive the real-time clock. In 32k-less mode, TCXO32K is used. While in 32k mode, only XOSC32K is used.
 - 32.768kHz low-speed external crystal (XOSC32K).
 - 32.768kHz low-speed internal clock fed by TCXO (TCXO32K).
- **Peripheral clock sources** — Three types of peripheral clock source options are used. Each peripheral has its own gating register:

Peripherals, such as EMI, SFC, ESC, SPI and SDIO have their own independent clock sources, LPOSC and UPLL, each having independent outputs enabling high flexibility. UPLL can generate independent clocks for USB.

The clock source of peripherals, such as DMA is the same as AHB/APB bus clock (F_{BUS}).

The clock source of low-speed peripherals, including I2C0, I2C1, I2C2, I2C_DMA, AUXADC and EFUSE is from general 26MHz MUX.

- **Clock-out capability:**

FREF. Outputs 26MHz TCXO clock by control GPIO.

CLKOUT. Outputs 32.768kHz clock based on 32k or 32k-less mode.

TCXO 26MHz is selected as a default system clock when powering up or resetting the chip. This clock source serves as an input to PLL to increase the CPU frequency (F_{CPU}) up to 530MHz when V_{CORE} is 0.8V. The application can then select the system clock as low-power RC oscillator (LPOSC) to decrease power consumption while frequency variation is acceptable. LPOSC can provide maximum of 384MHz clock.

Several prescalers can be used to configure the bus. The maximum frequency of the AHB1 and APB1 buses (F_{BUS}) is 265MHz when V_{CORE} is 0.83V. The low-speed APB2 bus domain is 26MHz. The frequency ratio of F_{CPU} and F_{BUS} needs to be 2:1:1. When V_{CORE} is 0.65V, the maximum frequency of the F_{CPU} is 192MHz and F_{BUS} are 96MHz, provided by LPOSC.

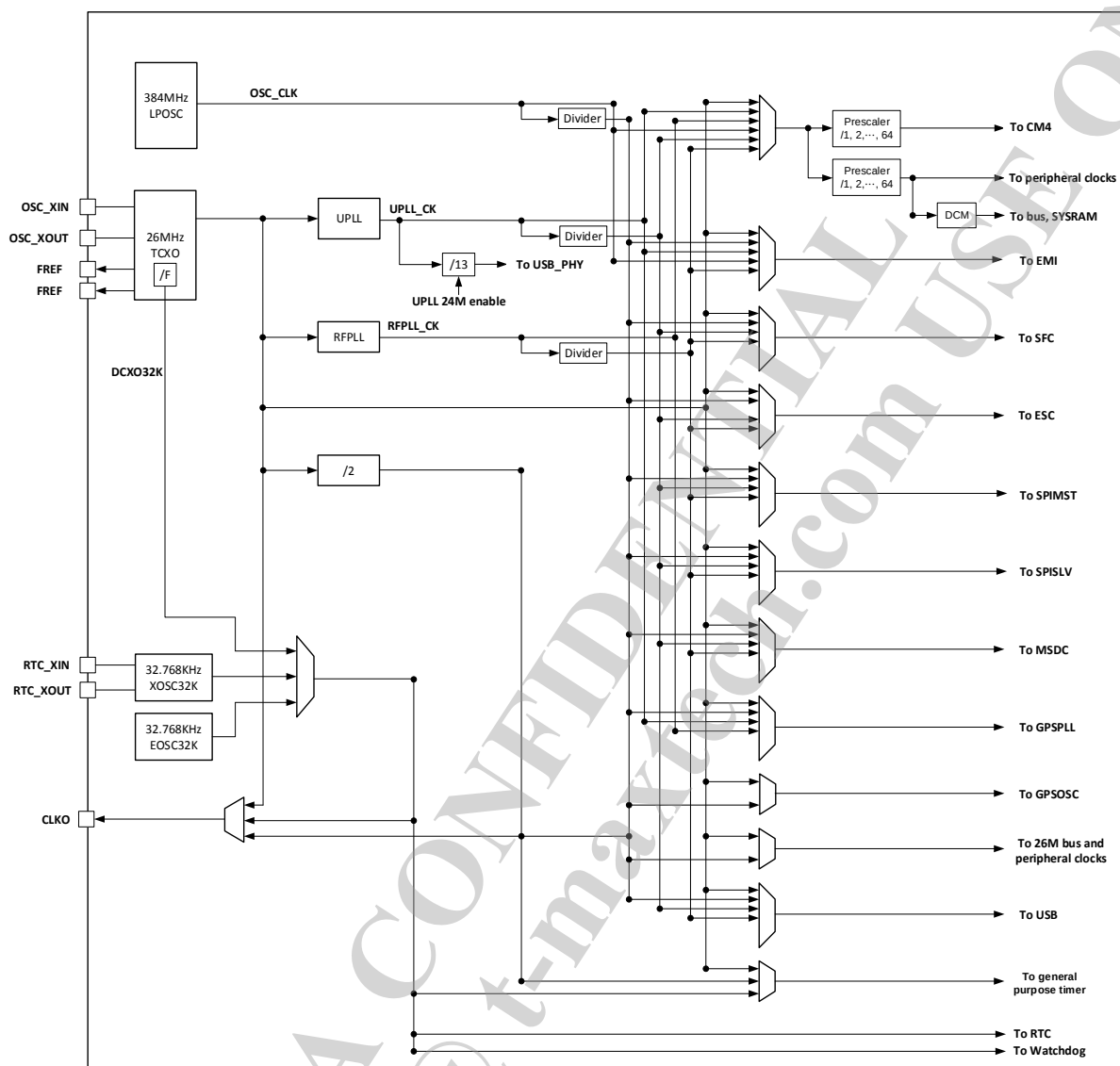


Figure 2.3-1. AG3335 clock source architecture

2.3.2 32kHz crystal oscillator (XOSC32)

The low-power 32kHz crystal oscillator XOSC32 is designed to work with an external piezoelectric 32.768kHz crystal and a load composed of two functional capacitors. It is designed to be a clock source of RTC for lower power platform. See Table 2.3-1 for the key performance.

The crystal parameters determine the oscillation allowance. Table 2.3-2 lists recommendations the best crystal parameters to be used with XOSC32.

Table 2.3-1. Functional specifications of XOSC32

Symbol	Parameter	Min.	Typical	Max.	Unit
VDIG18	RTC module power	-	1.8	-	V
Tosc	Start-up time	-	-	1	sec
Dcyc	Duty cycle	35	50	-	%
	Current consumption	-	1.8	-	μA
T	Operating temperature	-40	-	85	°C

Table 2.3-2. Recommended parameters for 32kHz crystal (XOSC32)

Symbol	Parameter	Min.	Typical	Max.	Unit
F	Frequency range	-	32768	-	Hz
GL	Drive level	-	-	0.5	μ W
$\Delta f/f$	Frequency tolerance	-	+/- 20	-	ppm
ESR	Series resistance	-	-	70	k Ω
C0	Static capacitance	-	0.9	-	pF
CL1	Load capacitance	-	12.5	-	pF

Under such CL range and crystal, the -R is more than 3 times bigger. If a larger CL is selected, the frequency accuracy will be decreased, and the -R will also degrade.

2.3.3 32kHz low-speed internal RC (EOSC32)

The low-power 32.768kHz embedded oscillator EOSC32 is designed to support function without 32.768kHz crystal. It can be configured by hardware option. When crystal is absent, RTC module has an embedded 32.768kHz oscillator to assist lifetime extension after master 32.768kHz clock is off. The key performance shown as bellow.

Table 2.3-10. Functional specifications of EOSC32

Symbol	Parameter	Min.	Typical	Max.	Unit
VDIG18	RTC module power	-	1.8	-	V
Fout	Output clock frequency	-	32.768	-	kHz
Dcyc	Duty cycle	20	50	80	%
	Current consumption	-	-	3	μ A
T	Operating temperature	-40	-	85	$^{\circ}$ C

2.3.4 Low-power oscillator (LPOSC)

2.3.4.1 Functional description

There is a low-power RC oscillator (LPOSC) generating 384MHz clock output with $\pm 4\%$ frequency variation. The benefit of a RC oscillator is that reference clock is not necessary. Moreover, this oscillator has a very low-power feature ($< 0.15\text{mA}$) and fast settling ($< 20\mu\text{s}$). To overcome the process variation, the software calibrates this OSC by frequency meter before using them.

2.3.4.2 Functional specifications

Table 2.3-3. LPOSC specifications

Symbol	Parameter	Min.	Typ.	Max.	Unit
Fout	Output clock frequency	-	384	-	MHz
	Settling time	-	20	-	μ s
	Output clock duty cycle	40	50	60	%
	Output clock jitter (period jitter)	-	125	-	ps

Symbol	Parameter	Min.	Typ.	Max.	Unit
DVDD	Digital power supply	0.62	0.65/0.7/0.83	0.88	V
AVDD	Analog power supply	1.62	1.8	1.98	V
T	Operating temperature	-40	-	125	°C
	Current consumption	-	0.15	-	mA
	Power-down current consumption	-	-	5	μA

2.3.5 Phase locked loop

2.3.5.1 Block description

There are two phase-locked loops (PLL) in the Phase-Locked Loop Block (PLLGP): UPLL and RFPLL.

Figure 2.3-2 shows the block diagrams of UPLL. The UPLL generates a 624MHz clock output, and the frequency dividers derive the fixed 312MHz and 24MHz clocks.

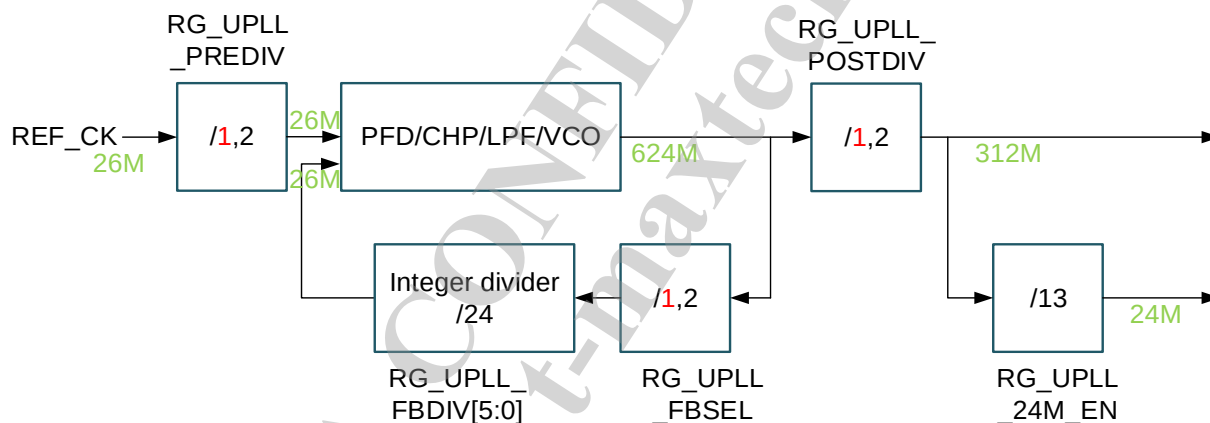


Figure 2.3-2. Block diagram of UPLL clock source

Figure 2.3-3 shows the block diagrams of RFPLL. The RFPLL generates a clock around 1570MHz.

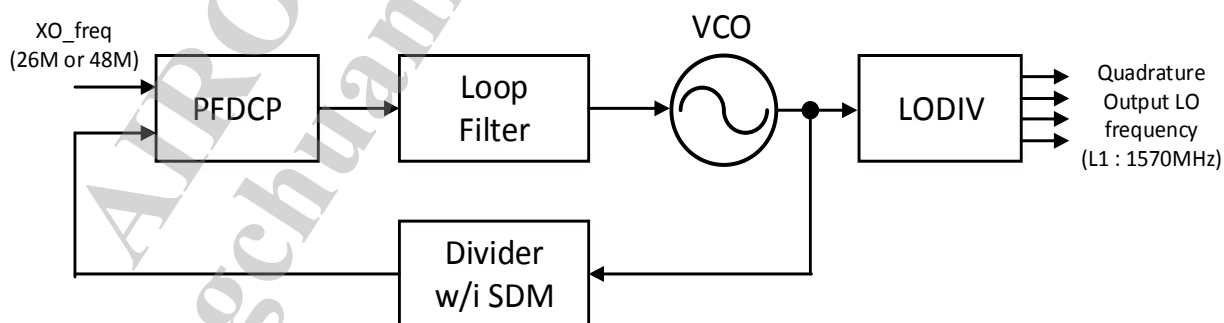


Figure 2.3-3. Block diagram of RFPLL clock source

These PLLs do not require off-chip components to operate, and can be turned off to save power.

After powering up, the PLLs are all off by default control register setting. The source clock signal for the PLLs is selected through multiplexers from 26MHz XTAL. The software maintains the PLL lock time while the clock selection is changing.

For power management, the MCU software configuration may stop MCU Clock by setting up the Sleep Control Register. Any interrupt requests to MCU can pause the sleep mode and return the MCU to the running mode.

2.3.5.2 Function specifications

The function specifications of PLLs are shown in the Tables below.

Table 2.3-4. UPLL specifications

Symbol	Parameter	Min.	Typ.	Max.	Unit
Fin	Input clock frequency	-	26/48	-	MHz
Fout	Output clock frequency	-	312	-	MHz
		-	24	-	
	Settling time	-	20	-	us
	Output clock duty cycle @ 312MHz	44	50	56	%
	Output clock duty cycle @ 24MHz	48	54	60	%
	Output clock jitter (rms period jitter)	-	±50 (312MHz)	-	ps
			±100 (24MHz)		ps
DVDD	Digital power supply	0.62	0.65/0.7/0.83	0.88	V
AVDD	Analog power supply	1.67	1.8	1.98	V
	(<100mVpp ripple @100kHz)				
	(<50mVpp ripple @500kHz)				
T	Operating temperature	-40	-	125	°C
	Current consumption	-	1.0	-	mA
	Power-down current consumption @ 1.8V	-	0.1	-	µA
	Power-down current consumption @ 0.7V, Typical with temperature of 27°C	-	0.4	-	µA

The function specifications of RFPLLs are shown in the Tables below.

Table 2.3-6. RFPLL specifications

Specifications of RFPLL				
Parameter	Min.	Typ.	Max.	Unit
VDD Supply	0.65	0.7	0.83	V
Current Consumptions	-	1.0	-	mA

Specifications of RFPLL				
Parameter	Min.	Typ.	Max.	Unit
Integrated RMS Phase Error (integrated Phase Noise from 1k~10MHz)	-	2.5	-	degree
Phase noise @ offset freq 120MHz	-	-145	-	dBc/Hz
Output Frequency (by mode)				
GPS / Galileo mode	-	1571.3	-	MHz
Glonass mode	-	1588.59	-	MHz
Beidou mode	-	1568.26	-	MHz
G3B mode	-	1581.94	-	MHz

2.4 Peripheral description

2.4.1 General purpose input/output (GPIO)

AG3335 platform offers many general purpose input/output (GPIO) pins. By setting up the control registers, the MCU software can control the direction and output value, as well as read the input value on these pins. The GPIOs are multiplexed with other functions to reduce pin count.

The GPIOs can be configured to send clock information out from the chip. There are four clock-out ports and each clock-out can be programmed to output the correct clock. In addition, when two GPIOs are multiplexed to the same peripheral IP input, the GPIO with a smaller serial number has a higher priority.

2.4.2 Direct memory access (DMA)

Two DMA controllers are placed on AHB bus to support fast data transfers and off-load the processor. With the controllers, specific devices on AHB or APB buses benefit greatly from quick completion of data movement from or to memory modules. The DMA controllers can also be used to connect two devices other than memory modules, as long as they can be addressed in memory space. Figure 2.4-1 illustrates the system connections.

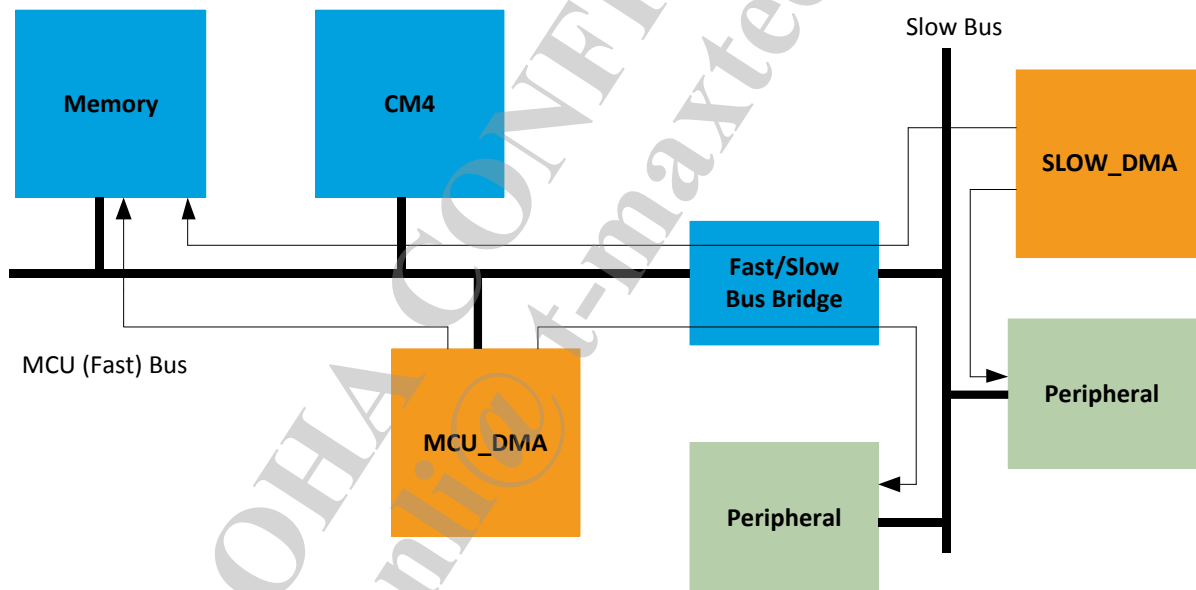


Figure 2.4-1. Variety data paths of DMA transfers

There is a round-robin arbitration mechanism that supports up to 5 channels in CM4_DMA and 12 channels in SLOW_DMA working simultaneously. Each channel has similar set of registers to be configured to different schemes as desired. Both interrupt and polling-based schemes are supported to manage the completion events.

The operating block diagram of the DMAs is shown in Figure 2.4-2.

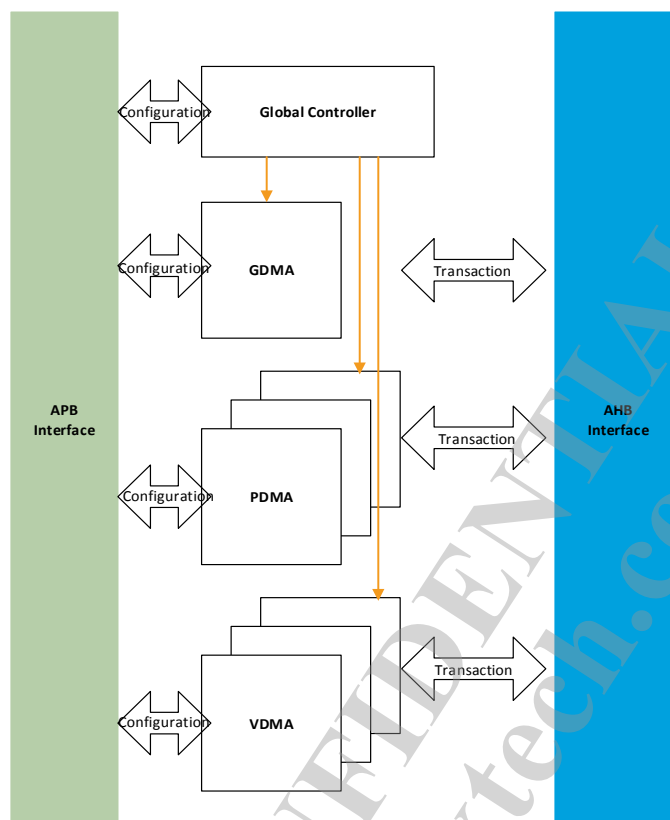


Figure 2.4-2. DMA block diagram

2.4.3 General purpose timer (GPT)

The general purpose timer (GPT) includes eight 32-bit timers. Each timer has four operation modes, ONE-SHOT, REPEAT, freerun with interrupt (FREERUN_I), and FREERUN. The GPT can operate on either system clock (13MHz) or real-time clock (RTC, 32.768KHz). The GPT is an always-on IP, which means it retains the previous configuration and runs even if the system goes into sleep mode. Note, in sleep mode, the clock source must be set to RTC because there is no system clock.

- **Timer mode**

Each GPT has four modes: ONE-SHOT; REPEAT; FREERUN_I; and FREERUN.

- ONE-SHOT mode – An interrupt occurs and the timer stops once GPT is timed out.
- REPEAT mode – An interrupt occurs and the timer resets once GPT is timed out.
- FREERUN_I mode – An interrupt occurs and the timer continues once the GPT is timed out.
- FREERUN mode – The GPT continues counting with no limit.

- **Timer clock source**

Each GPT can operate on either system clock or RTC. There is also a 4-bit clock divider, which divides the clock by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 16, 32, or 64. Note, that in sleep mode, only RTC is available.

- **Timer interrupt source**

All GPTs share a single interrupt source. The user can get the interrupt status by reading GPT_IRQSTA[11:0] and GPTx_IRQ_STA. To clear the interrupt, write 1 to GPTx_IRQ_ACK.

- **System wakeup source**

The interrupt signal is also connected to System Power Management (SPM) as a wakeup source.

2.4.4 True random number generator (TRNG)

The true random number generator (TRNG) is a device in power-down domain that generates random numbers from the ring oscillator (RO) outputs. Various types of ROs are adopted, including hybrid Fibonacci ring oscillator (H-FIRO), hybrid ring oscillator (H-RO), and hybrid Galois ring oscillator (H-GARO). An interrupt request (IRQ) is issued when the random data is successfully generated.

- Normal mode – turns on ROs and samples their output to generate random numbers. The IRQ is issued when the random data is valid.
- Freerun mode – continuously activates ROs to create interference on the power source. This can be used to help prevent side-channel attacks. The IRQ should be masked at this mode.

2.4.5 Real time clock (RTC)

The Real Time Clock (RTC) module provides time and data information and a 32.768kHz clock. The clock is selected from one of two clock sources: one is the external crystal oscillator (XOSC); the other is divided from TCXO.

When the AG3335 chipset is in RTC mode, a dedicated regulator supplies power to the RTC block. A low-power solution is provided to gate the 32.768kHz clock because RTC is one of the designs powered on when it is in RTC mode. To maintain RTC mode, we also allow RTC to use an alarm, a timer, and an external interrupt (EINT) to wake up the system. One dedicated pad is used to support the wakeup events.

2.4.6 Pulse width modulation (PWM)

Generic pulse width modulators (PWM) generate pulse sequences with a programmable frequency and a duty cycle for the LCD backlight. The duration of the PWM output signal is high if the internal counter value is between the threshold up and threshold down values. The waveform is shown in Figure 2.4-3.

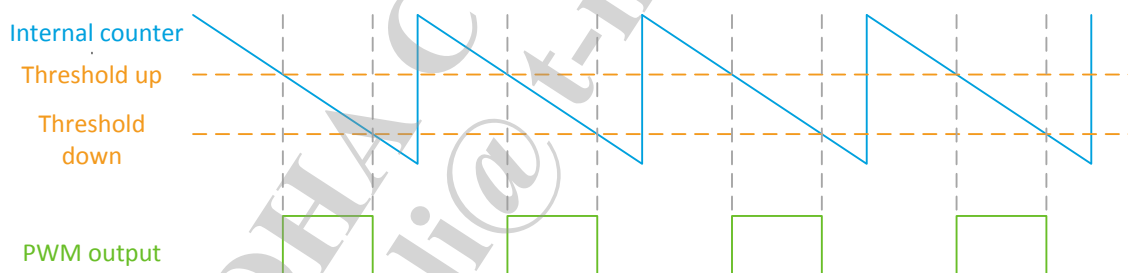


Figure 2.4-3. PWM waveform

- Classic mode – the output waveform is set by the internal counter, threshold up value, the threshold down value, clock source select and the clock prescaler scale.

2.4.7 eMMC/SD/SDIO card controller (MSDC)

The eMMC/SD/SDIO card host controller (MSDC) supports:

- 1) eMMC card specification version 4.41
- 2) SD memory card specification version 2.0
- 3) SDIO card specification version 2.0

The controller can be configured as a host for the eMMC/SD/SDIO card. There is one MSDC IP in this SOC.

The main features of the controller are:

- 32-bit access for control registers
- Built-in CRC circuit
- Support for PIO mode, basic DMA mode, descriptor DMA mode
- Interrupt capabilities
- Support for SD/SDIO speed mode:
 - Default Speed mode (DS)
 - High Speed mode (HS)
- Support for eMMC speed mode:
 - Backwards Compatibility with legacy MMC card (DS)
 - High Speed SDR mode (HS)
- Support for 1-bit/4-bit SD/SDIO/eMMC bus widths. The module is targeted at a 48MHz operating clock at 0.7V and 0.83V, and 26MHz at 0.65V. Data rates up to 48Mbps in 1-bit mode, 48 x 4 Mbps in 4-bit mode
- Programmable serial clock rate on SD/SDIO/eMMC bus (256 gears)
- Card detection capabilities; This SOC uses the EINT controller for card detection
- No support for SPI mode for SD memory card
- No support for suspend/resume for SDIO card

2.4.8 USB2.0 high-speed device controller (USB)

The USB2.0 controller supports both High-Speed(480Mbit/s) and Full-Speed(12Mbit/s). The controller is configured for supporting two endpoints to receive packets and four endpoints for sending packets (except for endpoint 0). These endpoints can be individually configured in the software to manage either bulk transfers, interrupt transfers, or isochronous transfers. There are four DMA channels and the embedded RAM size is configurable up to 3264 bytes. The embedded RAM can be dynamically configured to each endpoint.

Feature list	Description
Speed	High-Speed(480Mbit/s)/Full-Speed(12Mbit/s)
Enhanced feature	Generic Device
Endpoints	Four Tx; Two Rx
DMA channels	Four
Embedded RAM	3264 bytes

2.4.9 Inter-integrated circuit controller (I2C)

Inter-Integrated Circuit (I2C) is a two-wire serial interface. The two signals are SCL and SDA. SCL is a clock signal driven by the master. SDA is a bi-directional data signal that can be driven by either the master or the slave. This generic controller supports the master/slave role and conforms to the I2C specification.

- Master

- I2C compliant master mode operation
- Adjustable clock speed for Low-Speed and Fast-Speed mode operations
- Support for 7-bit/10-bit addressing
- Support for high-speed mode
- Support for the slave clock extension when in the open-drain mode
- START/STOP/REPEATED START conditions
- Polling/DMA transfer mode
- Multi-write per transfer (up to 65535 data bytes)
- Multi-read per transfer (up to 65535 data bytes)
- Multi-transfer per transaction (up to 65535 data bytes)
- Combined format transfer with the ability to change the length
- Slave
 - I2c compliant slave mode operation
 - Support for 7-bit addressing; 2-bit addressing can be configured by e-fuse
 - START/STOP condition
 - DMA transfer mode

2.4.10 Universal asynchronous receiver/transmitter (UART)

- The universal asynchronous receiver transmitter (UART) provides full-duplex serial communication channels between the baseband chipset and external devices.
- The UART has both M16C450 and M16550A modes of operation that are compatible with a range of standard software drivers. The extensions are designed to be software compatible with M16550A variants, but certain areas offer no consensus.
- The UART supports word lengths from five to eight bits, an optional parity bit, and one or two stop bits. The UAERT is completely programmable by an 8-bit CPU interface. A 16-bit programmable baud rate generator and an 8-bit scratch register are included with separate transmit and receive FIFOs. Two modem control lines and a diagnostic loop-back mode are provided. The UART also includes two DMA handshake lines. The DMA handshake lines show when the FIFOs are ready to transfer data to the CPU.
- Note that the UART is designed so that all internal operations are synchronized by the clock signal. This synchronization causes in minor timing differences between the UART and industry standard M16550A device, which means that the core is not clocked for clock identical to the original device.
- When the hardware resets, the UART goes into M16C450 mode. The related FIFOs can then be enabled and the UART can go into M16550A mode. The UART has other functions in addition to the M16550A mode. Each of the extended functions can be individually selected under software control.
- There are three UART channels: UART0; UART1; and UART2. Each channel has support for software flow control. All UART channels can support hardware flow control. Each UART has an individual interrupt source.
- For transmission, the UART supports word lengths from five to eight bits, with an optional parity bit and one or two stop bits.
- The UART supports standard baud rates of 110bps, 300bps, 1200bps, 2400bps, 4800bps, 9600bps, 19200bps, 38400bps, 57600bps, 115200bps, 230400bps, 460800bps, 921600bps, and non-standard baud rates from 110bps to 3Mbps.
- There are dedicated DMA channels for both transmit (TX) and receive (RX) for each UART.
- The UART supports automatic baud rate detection in RX mode. The recommended baud rate range is between 110bps and 115,200bps.

2.4.11 Serial peripheral interface (SPI)

The Serial Peripheral Interface (SPI) is a serial transmission protocol which supports single mode (four-pin), 3-wire mode (three-pin), dual mode (four-pin) and quad mode (six-pin) for increased data throughput. The maximum serial clock (SCK) frequency is 53MHz when the clock source is RFPLL, and 48MHz when the clock source is LPOSC. Note: single mode can support full duplex, but 3-wire, dual, and quad modes only support half duplex. Figure 2.4-4 is an example of the connection between the SPI master and SPI slave. Table 2.4-1 shows the characteristic of each pin.

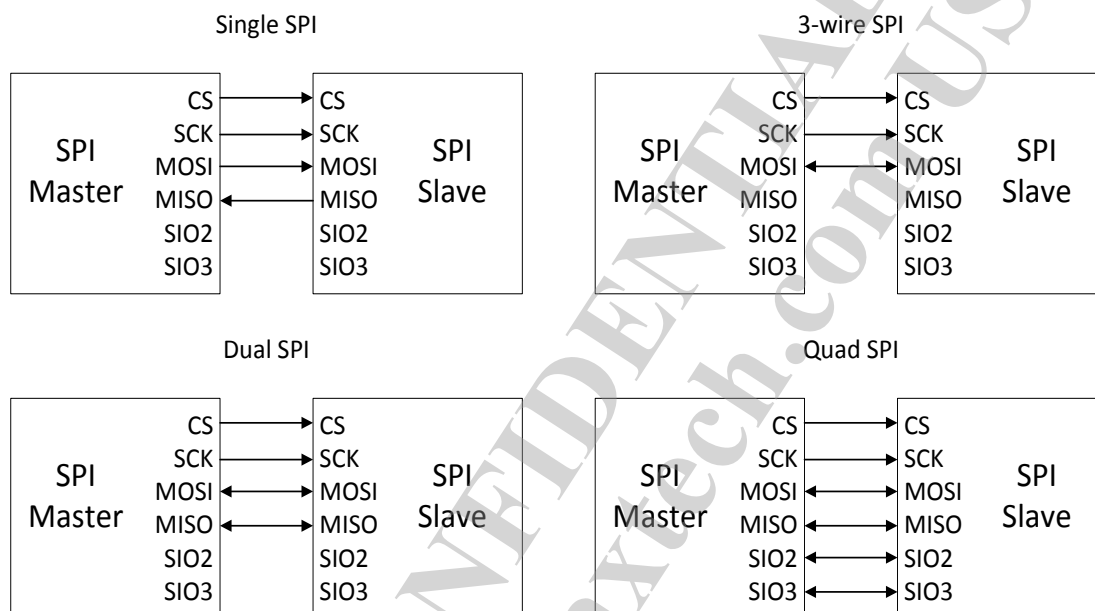


Figure 2.4-4. Pin connection between SPI master and SPI slave

Table 2.4-1. SPI master controller interface

Signal name	Type	Default value	Description
CS	O	1 (output)	Active low chip selection signal
SCK	O	0 (output)	The (bit) serial clock. Maximum SCK clock rate is 52MHz.
MOSI	I/O	1 (output)	Data signal 0
MISO	I/O	Pull down (input)	Data signal 1
SIO2	I/O	1 (output)	Data signal 2
SIO3	I/O	1 (output)	Data signal 3

2.4.11.1 SPI master controller

The SPI master controller supports single mode, 3-wire mode, dual mode, and quad mode. The controller can automatically change the port direction for data input/output according to registers **SPIM_TYPE** and **SPIM_RW_MODE**.

The SCK frequency can be configured as (source clock frequency)/N MHz, where N is in the range from 2 to 2¹⁷, with CPOL and CPHA features for different applications. CPOL defines the SCK polarity. CPHA defines the legal timing to sample data. The CS signal setup time, hold time, and idle time can also be configured. The detailed timing diagram of the SCK and CS is shown in Figure 2.4-5.

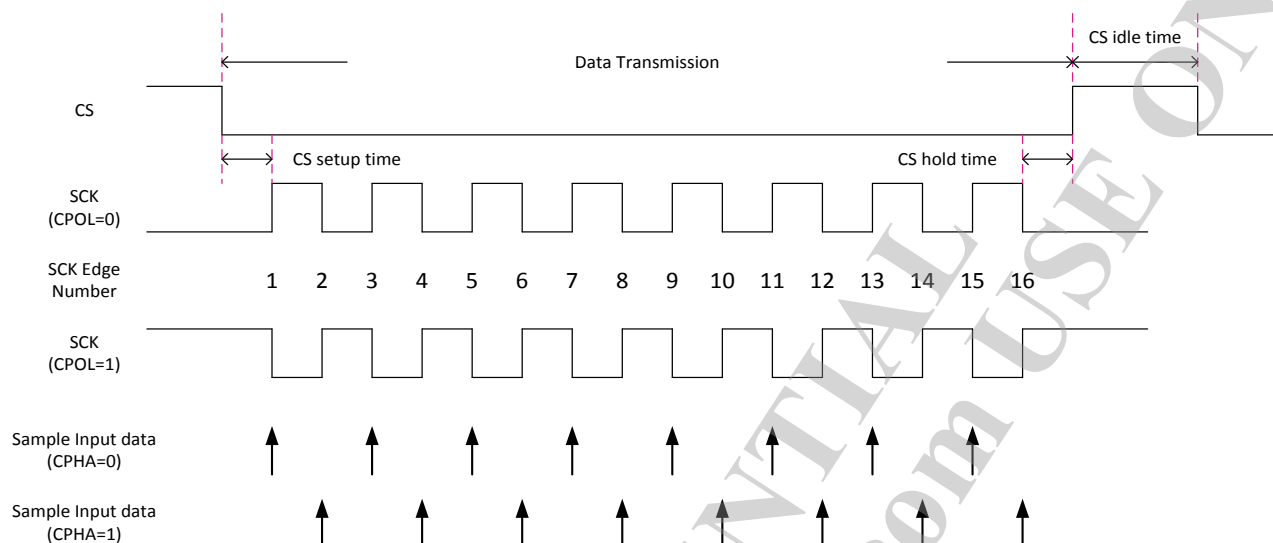


Figure 2.4-5. SPI transmission formats

There are two modes for data read/write in SPI master controller:

- Direct mode –The CPU directly writes data to or reads data from the SPI master controller FIFO.
- DMA (Direct Memory Access) mode – The SPI master controller includes DMA design, which can automatically and continuously write data from memory to the SPI master controller or read data from the SPI master controller to memory. In DMA mode, the endian of memory data is adjustable.

Unlimited length transmission can be achieved by enabling pause mode. In pause mode, the CS signal stays active after one transfer. During this period, the SPI master controller is in **PAUSE_IDLE** state and waits for the resume command to start the next transfer. Figure 2.4-6 is the state transition diagram.

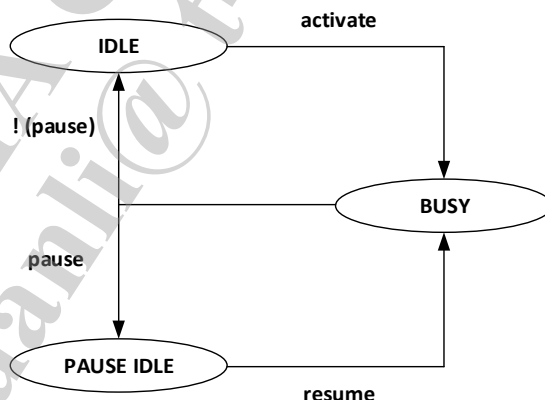


Figure 2.4-6. Operation flow with and without PAUSE mode

A configurable option to control CS de-assertion between byte transfers is available. The SPI master controller supports a special transmission format called CS de-assert mode. Figure 2.4-7 illustrates the waveform in this transmission format.

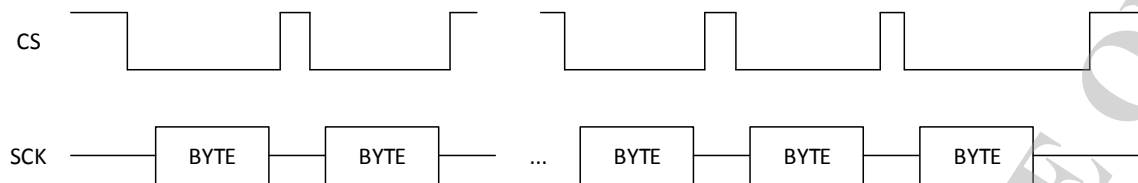


Figure 2.4-7. CS de-assert mode

When the SPI master controller operates in 3-wire, dual or quad mode, the transmission package includes three parts: command phase; dummy phase; and data phase.

- Command phase always operates in single mode.
- Dummy phase cannot transmit or receive data.
- Data phase operation depends on **SPIM_TYPE** and **SPIM_RW_MODE** settings. The command phase and dummy phase are useful for special applications, such as read or write serial flash data.

The sample clock, SCK and data delay is adjustable to solve the latency and skew issue.

If the critical path latency between master and slave is larger than half of SCK cycle, the SPI master controller may sample the wrong data. The critical path of SPI transmission includes two parts:

- Master transmits SCK to slave
- Slave feeds back data to master

The sampling clock delay (register **SPIM_GET_DELAY**) and sampling edge (register **SAMPLE_SEL**) can be adjusted to solve this issue. Each interval of **SPIM_GET_DLY** is 9.434 ns (RFPLL) or 10.416 ns (LPOSC). The detailed description is shown in Figure 2.4-8.

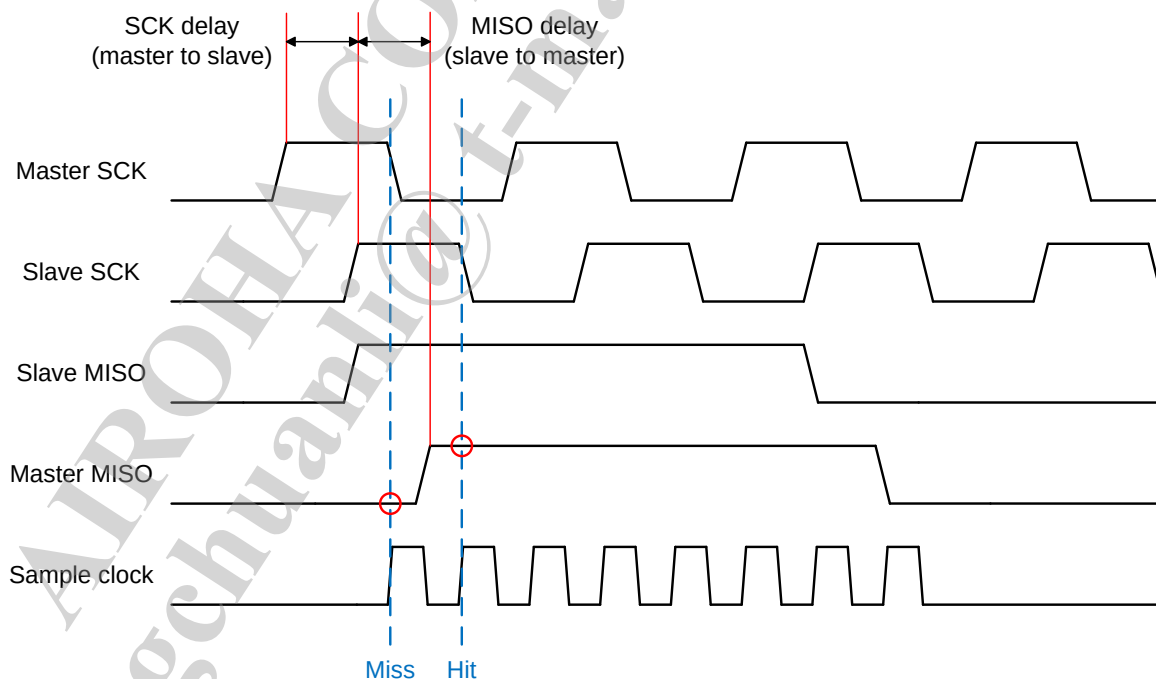


Figure 2.4-8. SPI master controller critical path sampling

If the timing skew between SCK and data is too big, the received data on slave can be corrupted. This issue can be solved by adjusting the delay on SCK and data path (registers **SPIM_SEL_ADDR** and **SPIM_SEL_WDATA**), as shown in Figure 2.4-9.

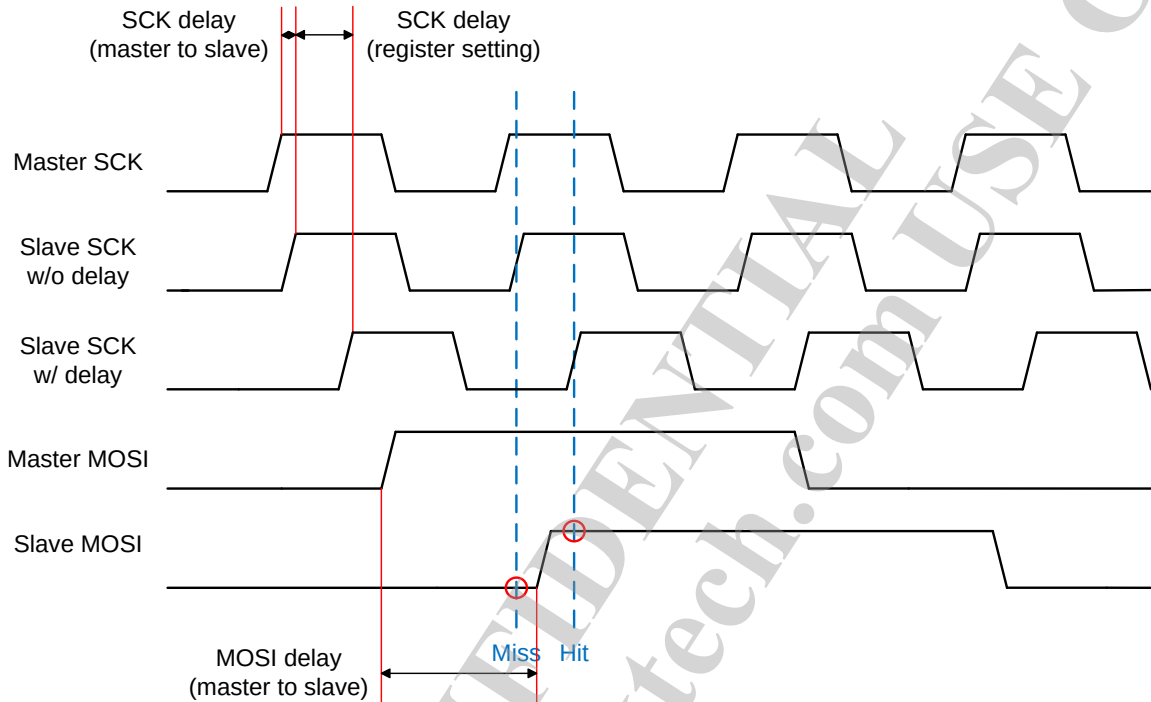


Figure 2.4-9. SPI master controller SCK and data delay

2.4.11.2 SPI slave controller

The SPI slave controller supports single mode, 3-wire mode, dual mode, and quad mode. The controller can automatically switch the port direction for data input/output according to **SPIS_TYPE**.

There are two methods for determining the memory address for the SPI slave to write/read data.

- When **SPIS_DEC_ADDR_EN** is 0, the address is determined by **SPISLV_BUFFER_BASE_ADDR** and CW/CR command from the master. The address and length from master CW/CR command is stored in **SPIS_TRANS_ADDR** and **SPIS_TRANS_LENGTH**. The start address of the transfer is "**SPISLV_BUFFER_BASE_ADDR + SPIS_TRANS_ADDR**", and the end address is "**SPISLV_BUFFER_BASE_ADDR + SPIS_TRANS_ADDR + SPIS_TRANS_LENGTH**". The CW/CR command can only succeed if the end address does not exceed the maximum available memory address, which is "**SPISLV_BUFFER_BASE_ADDR + SPISLV_BUFFER_SIZE**".
- When **SPIS_DEC_ADDR_EN** is 1, the address is directly determined by **SPISLV_BUFFER_BASE_ADDR**.

The maximum SCK frequency supported is 52 MHz with CPOL and CPHA features. CPOL defines the SCK polarity. CPHA defines the legal timing to sample data. The detailed timing diagram of the SCK and CS is shown in Figure 2.4-5.

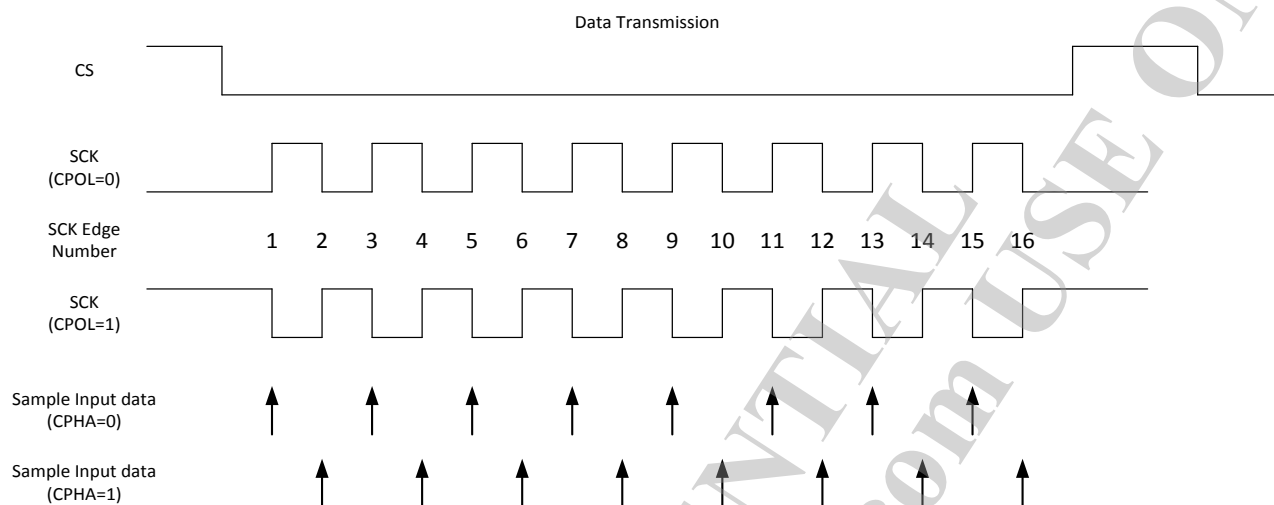


Figure 2.4-10. SPI transmission formats

The SPI slave controller supports the early transmission feature to solve the data path latency issue. The timing diagram is shown in Figure 2.4-11.

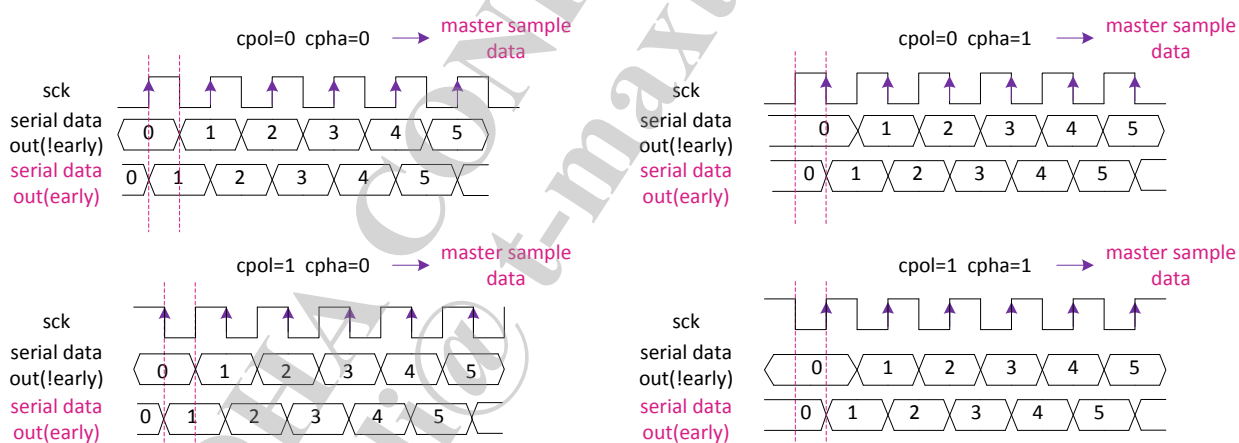


Figure 2.4-11. SPI slave controller early transmission

2.4.12 Serial flash controller (SFC)

The serial flash controller (SFC) supports convenient high-speed serial NOR flash access and flexible macro command sequence read/write.

- High-speed serial NOR flash access
- Flexible macro serial NOR flash command sequence read and write
- Supports 1.8V serial flash memory
- Supports 26, 62.4, 78 and 104MHz for serial flash device
- Supports serial flash SPI, QUAD I/O and QPI mode
- Supports serial flash memory with 3-byte and 4-byte addresses

2.4.13 External SRAM Controller (ESC)

The external SRAM controller (ESC) provides the platform with the capability to extend its memory space with external SPI SRAM/Flash. There is also support for the 1IO/4IO/QIO protocol access.

ESC supports up to 26MHz for the SPI protocol SRAM/SF device in the project. The bus master can use direct mode or macro mode to access SRAM/SF. Macro mode is flexible and allows the user to issue the SRAM/SF command. ESC supports SRAM/SF with a 3-byte address. ESC also supports SRAM/SF 1 IO, 4 IO and QIO modes.

- 1IO mode: command, addr and data through 1 bit IO
- 4IO mode: command 1 bit; addr and data through 4bits
- QIO mode: command, addr and data all through 4bits

The maximum serial SRAM clock frequency is 26MHz.

2.4.13.1 Direct read

Direct read mode provides a convenient way to read data programmed in SRAM/SF for the Cortex-M4, and peripherals on AHB.

First, the master sends an AHB read to ESC to activate Direct Read Engine to generate an SPI command or address according to the register settings or bus signals.

After sending an SRAM/SF command or address to the device, ESC latches SRAM/SF read data and puts it on the AHB data bus for the master. In short, direct read mode converts an AHB read to an SRAM/SF read command and returns SRAM/SF contents through AHB. If other SRAM/SF commands are required, it should be done using macro read and write modes.

2.4.13.2 Direct write

Direct write mode provides a way to write data to SRAM for CM4 and peripherals on AHB. Master set AHB configuration (e.g., wrap_en, critical first...), set SRAM/SF mode (1IO,QIO) by macro write, set ESC mode (1IO, QIO), set timing between ESC & SRAM/SF (e.g., time from cs = 0 to first clock, cs = 1 time between 2 R/W sequence). Master then does the direct-write by accessing the AHB address (ESC_MEM_BASE + offset).

2.4.13.3 Macro read and write

Macro read and write is designed for other SRAM/SF operations, including read ID, program, erase, and more. The user should write the SRAM/SF command (including opcode, address and data) to GPRAM in their code. When the macro read/write engine receives a trigger, it reads all of the user-defined SRAM/SF commands in GPRAM and sends them to SRAM/SF. The results of SRAM/SF command execution are latched and written to GPRAM for the user.

2.4.13.4 Masters arbitration

Masters access through same bus – wait last SPI done then next bus granted master

Masters access through different bus – macro mode will execute after direct mode state becomes idle

Master access ESC by direct mode when ESC macro mode is not idle

When AHB req mask (AHB_REQ_MASK. AHB_REQ_MASK) = 0, AHB suspends, AHB request is pended, data is correct.

When AHB req mask = 1, AHB does not suspend, AHB request is aborted, data is wrong. ESC issues an interrupt to indicate the event.

2.5 Analog baseband

2.5.1 Auxiliary ADC (AUXADC)

2.5.1.1 Block description

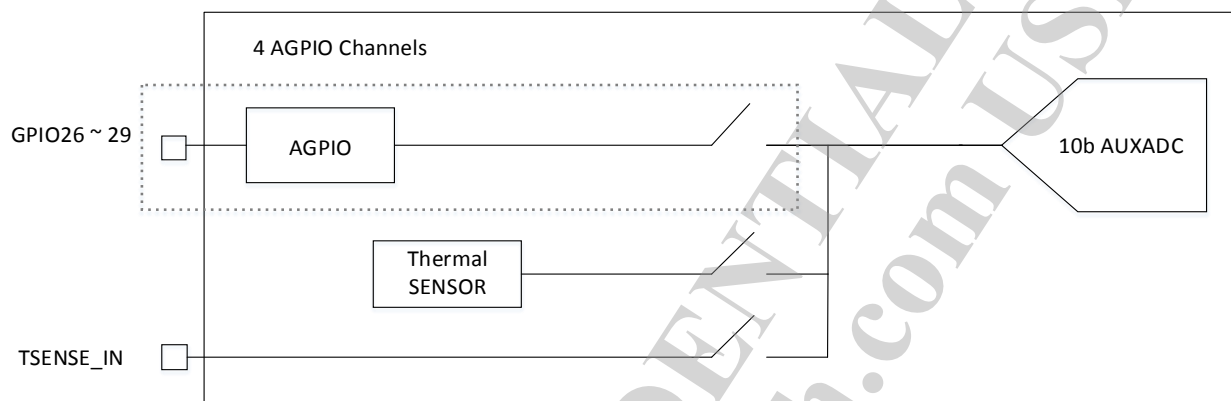


Figure 2.5-1. Block diagram

The auxiliary ADC includes the following functional blocks:

- 1) Analog multiplexer. Selects signal from one of the six auxiliary input pins. Real-world messages, such as temperature, are monitored and translated to the voltage domain.
- 2) 10-bit A/D converter: Converts the multiplexed input signal to 10-bit digital data.
- 3) Internal reference voltage generator supply 1.8V reference voltage.

Table 2.5-1. Auxiliary ADC input channels

Channel	Application	Input range [V]
0	GPIO29	0V to 1.8V
1	GPIO26	0V to 1.8V
2	GPIO28	0V to 1.8V
3	GPIO27	0V to 1.8V
4	Internal Thermal sensor	0V to 1.8V
5	External Thermal sensor	0V to 1.8V

2.5.1.2 Functional specifications

The functional specifications of the auxiliary ADC are listed in Table 2.5-2.

Table 2.5-2. Auxiliary ADC Specifications

Symbol	Parameter	Min.	Typ.	Max.	Unit
N	Resolution	-	10	-	Bit
FC	Clock rate	-	3.25	-	MHz
FS	Sampling rate at N-Bit	-	FC/N+4	-	MSPS
VIN	Input swing	0.00	-	1.8	V

Symbol	Parameter	Min.	Typ.	Max.	Unit
	Clock latency	-	N+4	-	1/FC
DNL	Differential nonlinearity	-	± 1	-	LSB
INL	Integral nonlinearity	-	± 2	-	LSB
OE	Offset error	-	± 10	-	mV
FSE	Full swing error	-	± 10	-	mV
DVDD	Digital power supply	0.65	0.7	0.83	V
AVDD	Analog power supply	1.62	1.8	1.98	V
Temp	Operating temperature	-40	-	85	°C
	Current consumption				
	Power-up	-	300	-	μA
	Power-down	-	1	-	μA

To guarantee the accuracy, ADC's output value is calculated by the following equation

Output value = ((code/4096-OE)*1.4/(1+GE)), where GE and OE are gain-error and offset-error preset in e-Fuse. Please refer to reference manual for detail e-Fuse address information.

3 GPSSYS

3.1 Abbreviations

Table 3.1-1 shows the abbreviations used in this document.

Table 3.1-1. Abbreviations

Abbreviation	Definition
GNSS	Global Navigation satellite system
AIC	Active interference cancellation
PPS	Pulse per second

3.2 Overview

AG3335 contains a high-performance multi-GNSS solution, including on-chip RF controller, digital baseband and a processing engine DSP. It is able to achieve the industry's highest level of sensitivity, accuracy and Time-to-First-Fix (TTFF) with the lowest power consumption.

3.3 Features

AG3335 supports the following GNSS signals:

- L1 band: GPS/GLONASS/GALILEO/BEIDOU, also supports QZSS/SBAS
- L5 band: GPS/GALILEO/BEIDOU/IRNSS

Besides, AG3335 provides the following services:

- Support WAAS/EGNOS/MSAS/GAGAN
- 12 multi-tone AIC in both L1 and L5 band for removing unwanted signals
- Indoor and outdoor multi-path detection and compensation
- RTCM ready
- PPS GPS time reference with adjustable duty cycle and ± 2 ns accuracy as select GPIO10(mode:4, PPS0) and GPIO11(mode:4, PPS1)
- FCC E911 compliance and A-GPS
- Max fixed update rate up to 10Hz
- Advanced software feature:
 - Periodic mode/GLP mode for lower power consumption
 - EPO™ orbit prediction
 - EASY™ self-generated orbit prediction
 - LOCUS™ logger function

3.4 Block diagram

The block diagram for the multi-GNSS receiver is shown in Figure 3.4-1.

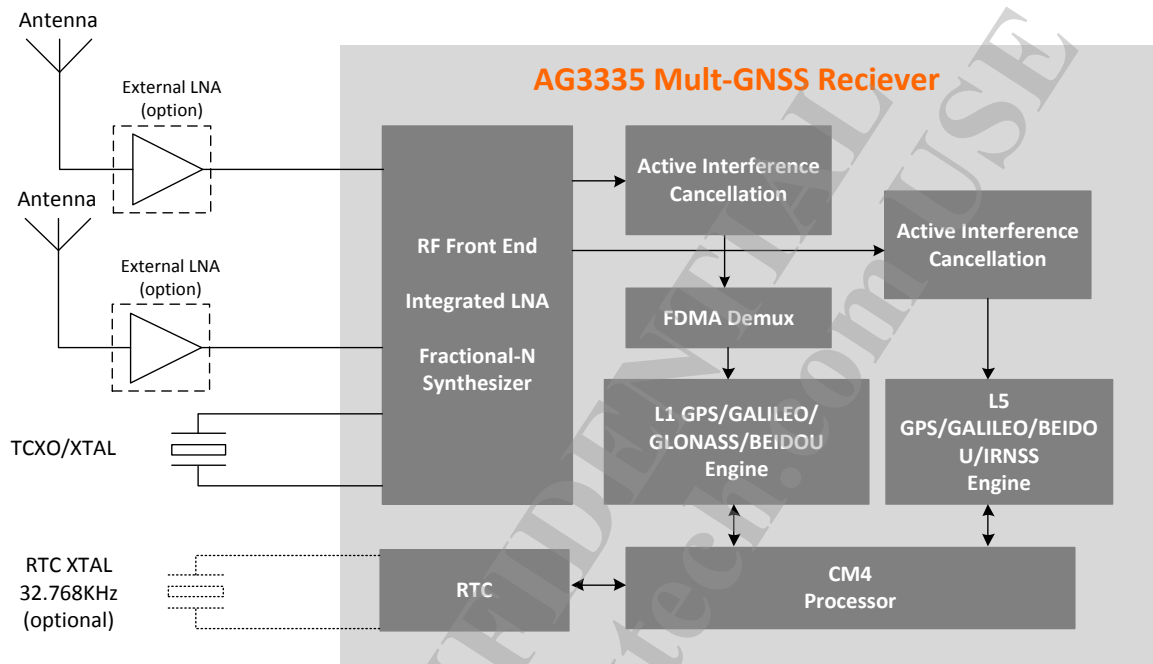


Figure 3.4-1. AG3335 Multi-GNSS receiver block diagram

4 RF Subsystem Features

4.1 Functional Block Diagram

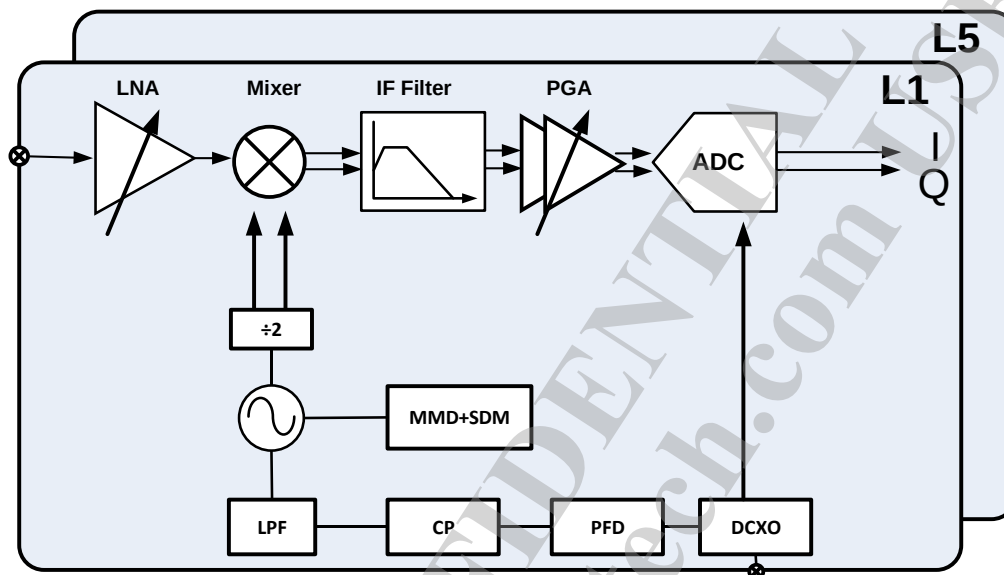


Figure 4.1-1. AG3335 RF functional block diagram

4.2 Low Noise Amplifier (LNA) and Mixer

An external antenna and high gain external LNA connected to the internal LNA in low-gain mode, which offers high linearity. In this configuration, the external LNA gain ranging from 0 to 36 dB is recommended.

The mixer down converts the amplified L1 band (1575.42 MHz) signal to a 4.092 MHz differential IF signal. The current chip provides three configurations to choose from — high-gain LNA, mid-gain LNA and low-gain LNA. The high-gain LNA is used for low-cost solution without external LNA. The low-gain LNA offers extremely low RF current consumption but the worst noise figure performance. The down-conversion mixer is a single-ended passive mixer with current mode interface between the mixer and multi-modes low pass filter.

4.3 Voltage Controlled Oscillator (VCO) and synthesizer

The frequency synthesizer includes a crystal oscillator, VCO, divider, phase frequency detector (PFD), charge pump (CP) and loop filter, which are all integrated on chip. The VCO is auto-calibrated to its required sub-band when the chip is powered on. The synthesizer has two topologies — integer-N and fractional-N, selectable through the baseband control. Integer-N synthesizer only supports 16.368 MHz. Other clock modes in a range from 12.6 MHz up to 40 MHz are supported by fractional-N synthesizer, together with a sigma-delta modulator (SDM) and multi-modulus divider (MMD).

4.4 Low Pass Filter (LPF)

Support the current-mode LPF provides modes for the various combinations of GNSS constellations.

4.5 Programmable Gain Amplifier (PGA)

The PGA has approximately 20 dB of gain control range with approximately 2 dB per step. The maximum gain

is around 20 dB. High-pass filter (HPF) circuits are implemented among PGAs to remove DC offset quickly.

4.6 Analog-to-Digital Converter (ADC)

The differential IF signal is quantized by a high performance ADC. The ADC sampling clock is provided by the divided clock from the local oscillator (LO).

4.7 Active Interference Canceller (AIC)

The AIC can detect 12 different single-tone interference signals. It can then track the phase and frequency of these 12 interference signals to provide continuous cancellation.

4.8 RF subsystem electrical characteristics

4.8.1 DC electrical characteristics

Parameter	Parameter	Min.	Typ.	Max.	Unit
I _{cc}	Total supply current:L1 band + L5 band (Total supply current = RX + SX + LDO)	-	7	-	mA
I _{cc} (L1 GPS)	Total supply current RF L1 GPS Ultra low power mode		2		mA

4.8.2 RX chain from LNA to PGA, before ADC

Parameter	Condition	Min.	Typ.	Max.	Unit
Noise figure	SOC on: High gain LNA SOC on: Low gain LNA	-	3 8	-	dB
Image rejection ratio			30		dB
V _{cc}		0.6	0.66	0.8	V

4.8.3 Receiver front-end part (LNA only)

Parameter	Condition	Min.	Typ.	Max.	Unit
RF input frequency		-	1.57542	-	GHz
Lo frequency			1.8175		GHz
Input return loss			-10		dBm
Voltage gain -- Av	High gain LNA Low gain LNA		60 45		dB
Noise figure	High gain LNA Low gain LNA		2.5 6		dB

4.8.4 Mixer and low pass filter (LPF)

Parameter	Condition	Min.	Typ.	Max.	Unit
Filter type	5 th order butterworth low pass filter				
Voltage	Supply voltage	0.6	0.66	0.8	V

Parameter	Condition	Min.	Typ.	Max.	Unit
BW _{3dB}	3dB bandwidth		2.5/27		MHz
Filter frequency response (2.5M/27M)	Rejection band attenuation at f = 3MHz f = 10MHz f = 15MHz f > 20MHz		23/12 54/45 65/54 72/60		dB
Voltage gain -- Av	High gain LNA Low gain LNA		32 20		dB

4.8.5 Programmable gain amplifier (PGA)

Parameter	Condition	Min.	Typ.	Max.	Unit
Voltage	Supply voltage	0.6	0.66	0.8	V
BW _{3dB}	3dB bandwidth		2.5/27		MHz
Voltage gain -- Av	Gain range	0		20	dB
Gain step	Gain step (5 bits)		2		dB

4.8.6 9-bit quantizer (ADC)

Parameter	Condition	Min.	Typ.	Max.	Unit
Voltage	Supply voltage	0.6	0.66	0.8	V
Input sampling clock	Operating frequency		83		MHz
Input signal frequency	Input signal center frequency		4.092		MHz
Resolution			9		bits

4.8.7 Integrated synthesizer

Parameter	Condition	Min.	Typ.	Max.	Unit
Fosc	VCO oscillation frequency		3163.5		MHz
V	Tuning voltage range	0.2		V _{cc} -0.2	V
DIV	Programmable divider ratio	32		127	
T _{start}	Circuit start-up time			100	us

5 Power Management Unit and Low-Power Control System

5.1 Overview

The power management unit (PMU) manages the power supply of the entire chip, including the processor, digital logic, memory, RF, oscillator, and more. There are two power input sources for AG3335:

- 1) For PMU control and RTC timer control (AVDD18_VRTC_IN, VDD_VSYS_IN, AVDD18_IN) – The PMU contains an Under-Voltage Lockout (UVLO) circuit, several low drop-out regulators (LDOs), several power switches, and a reference band-gap circuit. The circuits are optimized for low quiescent current, low drop-out voltage, efficient line/load regulation, and low output noise. It also contains the real time clock control and alarm logic, and to provide AG3335 with low-power system control. The voltage ranges for AVDD18_VRTC_IN, VDD_VSYS_IN and AVDD18_IN are from 1.62V to 1.98V, 1.15V to 1.98V, and 1.7V to 1.9V respectively.
- 2) For RF control (VDD_RF_IN) – operated by a wider input voltage range (0.9V to 1.98V) and contains the Low Drop-out Regulators (LDO) with low output noise for RF circuit to maintain RF performance.

5.2 Low-power operating mode

There are four low-power operating modes: ACTIVE, SLEEP, RTC, and OFF. The power state diagram for AG3335 is shown in Figure 5.2-1.

- In **ACTIVE** mode, the Cortex-M4 and GPSSYS power states operate independently, and both have Active and Sleep modes. For Cortex-M4, it can also go into Idle mode if a faster resume latency is necessary for different applications.
- When the Cortex-M4 and GPSSYS are in sleep mode, the chipset goes into **SLEEP** mode. The PMU can be changed to low-power mode during Sleep mode to conserve power.
- **RTC** mode provides a lower current consumption than SLEEP mode. It is suitable for applications that remain idle for a long period. Triggering RTC mode is software configurable. The RTC timer or EINT can be used to exit RTC mode.
- **OFF** mode is controlled by the CHIP_EN pad. Only the always-on PMU logics are active to maintain the lowest current consumption during OFF mode.

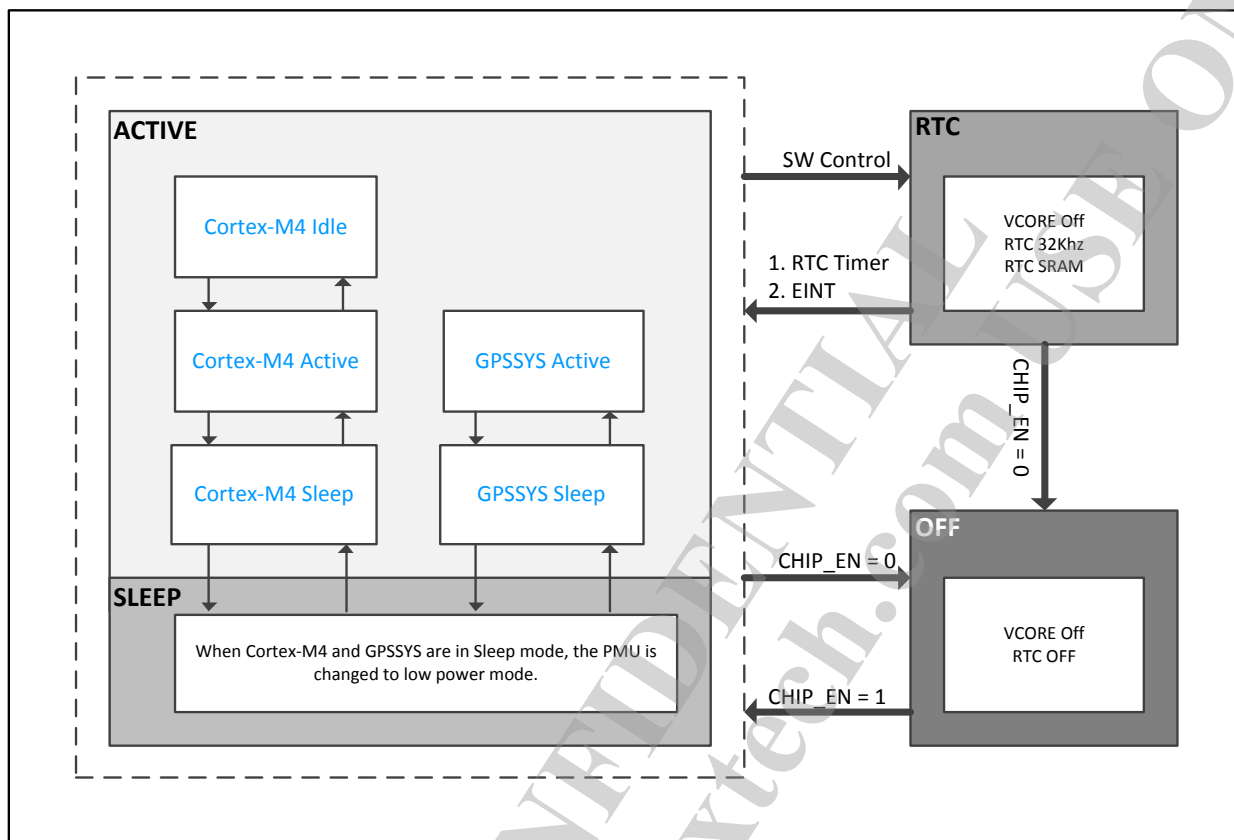


Figure 5.2-1. AG3335 Cortex-M4 and GPSSYS power state and power mode

5.3 PMU architecture

The PMU power grid is shown in Figure 5.3-1. It contains five LDOs and three power switches (PSWs). Table 3.1-1 shows the LDO output range, default voltage setting, and the output current capabilities. The LDOs are VRTC, VCORE, VSRAM, VRF and VUSB, which are used for the RTC macro, digital core, SRAM, RF circuits, and USB circuit. PSW_VIO18 is used for IOs, PLL, LPOSC, and internal memory devices. PSW_TCXO and PSW_USB18 provide power for TCXO and the USB circuit.

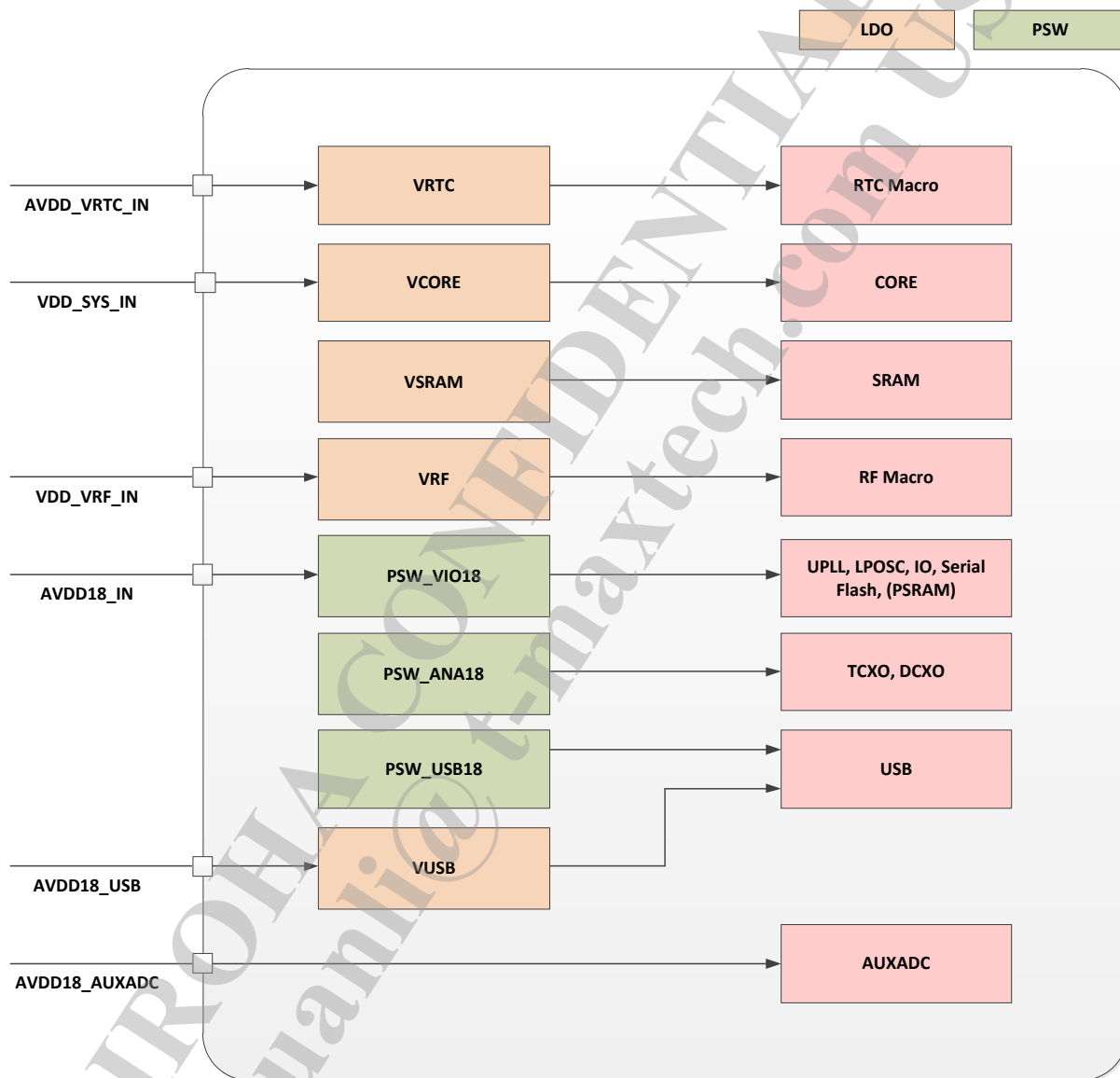


Figure 5.3-1. AG3335 power grid

Table 5.3-1. LDO characteristics

TYPE	Name	Output Voltage Range(V)	Default on Voltage (V)
LDO	VRTC	• 0.7~0.8	0.75
	VCORE	• 0.55~0.83	0.7
	VSRAM	• 0.6~0.925	0.9
	VRF	• 0.6~0.7	Default off
	VUSB	• 1.22	Default off
PSW	PSW_VIO18	• 1.8	1.8
	PSW_TCXO	• 1.8	1.8
	PSW_USB18	• 1.8	Default off

Table 5.3-1 lists the internal BUCK/ LDO output range, default voltage setting, and the output current capabilities.

5.4 Power on/off sequence

The PMU power on/off sequence is shown in Figure 5.4-1. The AVDD18_VRTC_IN, AVDD18_IN, VDD_SYS_IN, VDD_VRF_IN, AVDD18_AUXADC come from external power source. When these power are ready and CHIP_EN is high, the internal power source, such as VRTC, VCORE, VSRAM and VIO18, AVDD18_TCXO, will be turn on sequentially. The power off sequence is triggered by SW and occurs in the reverse order of the power on sequence. The extra control pin, EXT_PWR_EN, is an SW control output pin to control external power source for AVDD18_IN, VDD_SYS_IN, and VDD_VRF_IN. It can reduce PMIC IQ in RTC mode. Figure 5.4-2 also shows sequence for the entry and exit of RTC mode.

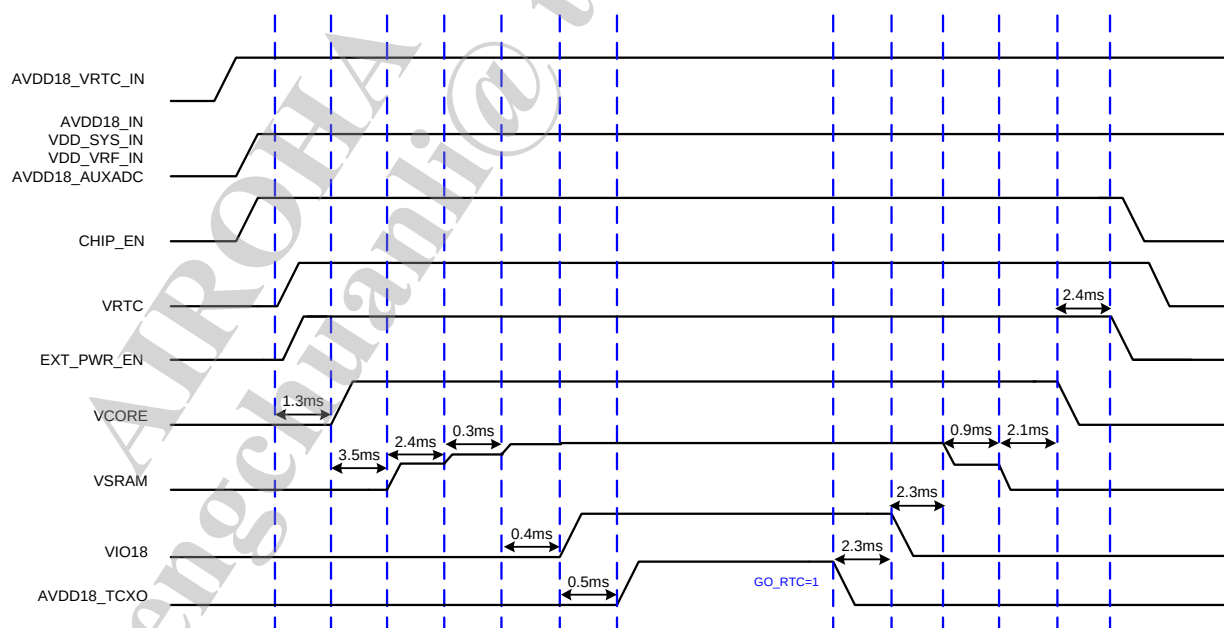


Figure 5.4-1. AG3335 power on/off sequence

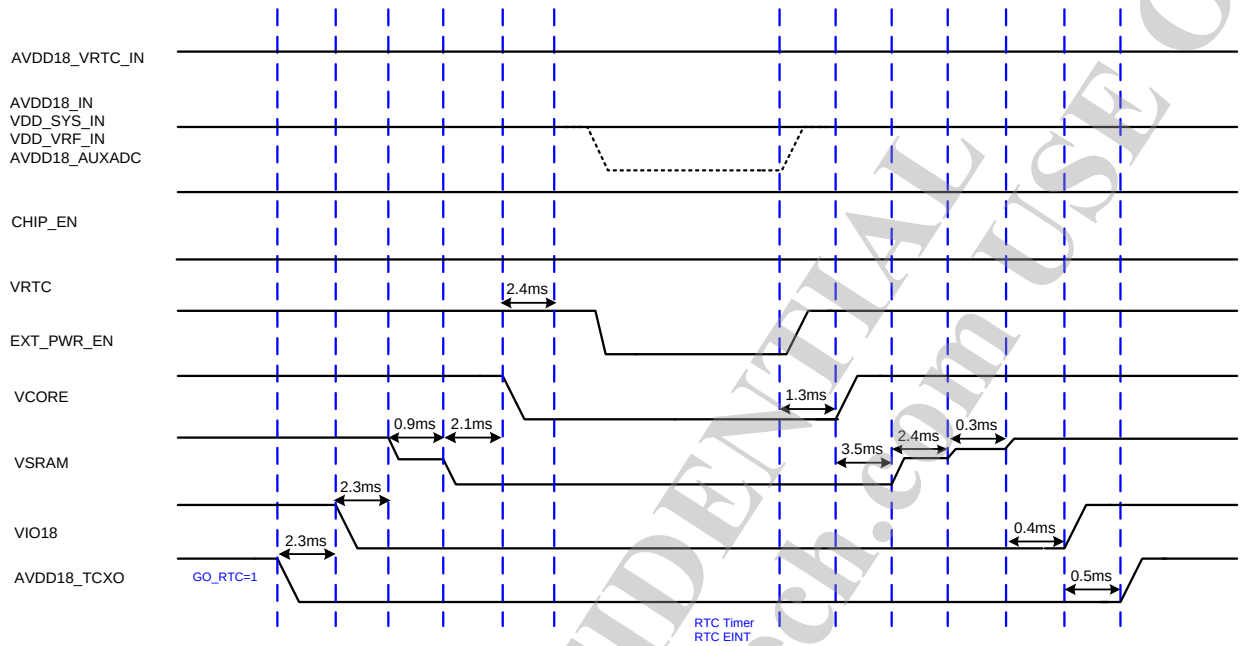


Figure 5.4-2. AG3335 power on/off sequence for RTC mode

6 Pin Description

6.1 AG3335A

6.1.1 AG3335A ball diagram

For AG3335A, a TFBGA 4.2mm*4.2mm, 83-ball, 0.4mm pitch package is offered. Pin-outs and the top view for this package are shown in Figure 6.1-1.

	1	2	3	4	5	6	7	8	9	10	84
A	AVSS_RF_L1_L5	RF_IN_L1	RF_IN_L5		RTC_EINT	AVDD18_IN		VIO18	VCORE	AVDD18_AUXADC	A
B	AVDD07_LO	AVDD07_L1_L5	AVSS_RF_L1_L5	AVSS_RF_L1_L5	VRF	AVDD18_VRTC_IN	VDD_VRF_IN	TESTMODE	AVSS_AUXADC	TSENSE_IN	B
C	OSC_XIN	AVDD07_ANA	AVSS_RF_ANA	CHIP_EN	FREF0	AVSS18_VSYS	VDD_SYS_IN	AVDD18_TCXO	EXT_PWR_EN	VRTC	C
D		OSC_XOUT	AVDD07_DCXO	AVSS_RF_DCXO			FSOURCE_D	RTC_XIN	RTC_XOUT		D
E	GPIO29	GPIO26		FREF1			GNDK		AVDD18_USB	USB20_DM	E
F	GPIO28	GPIO27	GPIO36	GPIO25	GNDK		VCKK		VUSB	USB20_DP	F
G		GPIO35	GPIO34	GPIO30	GPIO16	GPIO12	GPIO7	VSRAM	GPIO0		G
H	GPIO32	GPIO33	GNDK	GPIO19		GPIO13	GPIO9	GPIO8	GPIO1	GPIO2	H
J	GPIO31	GPIO24	GPIO21	GPIO20	GPIO18	GPIO14	GPIO10	GPIO11	GPIO4	GPIO3	J
K	DVDD18_IO	GPIO23	GPIO22		GPIO17	GPIO15		GPIO6	GPIO5	DVDD_MCO	K
	1	2	3	4	5	6	7	8	9	10	

Figure 6.1-1. AG3335A ball diagram and top view

6.1.1.1 AG3335A pin coordination

Table 6.1-1. AG3335A pin coordinates

Pin#	NETNAME	Pin#	NETNAME	Pin#	NETNAME
A1	AVSS_RF_L1_L5	B1	AVDD07_LO	C1	OSC_XIN
E1	GPIO29	F1	GPIO28	H1	GPIO32

Pin#	NETNAME	Pin#	NETNAME	Pin#	NETNAME
J1	GPIO31	K1	DVDD18_IO	A2	RF_IN_L1
B2	AVDD07_L1_L5	C2	AVDD07_ANA	D2	OSC_XOUT
E2	GPIO26	F2	GPIO27	G2	GPIO35
H2	GPIO33	J2	GPIO24	K2	GPIO23
A3	RF_IN_L5	B3	AVSS_RF_L1_L5	C3	AVSS_RF_ANA
D3	AVDD07_DCXO	F3	GPIO36	G3	GPIO34
H3	GNDK	J3	GPIO21	K3	GPIO22
B4	AVSS_RF_L1_L5	C4	CHIP_EN	D4	AVSS_RF_DCXO
E4	FREF1	F4	GPIO25	G4	GPIO30
H4	GPIO19	J4	GPIO20	A5	RTC_EINT
B5	VRF	C5	FREF0	F5	GNDK
G5	GPIO16	J5	GPIO18	K5	GPIO17
A6	AVDD18_IN	B6	AVDD18_VRTC_IN	C6	AVSS18_VSYS
G6	GPIO12	H6	GPIO13	J6	GPIO14
K6	GPIO15	B7	VDD_VRF_IN	C7	VDD_SYS_IN
D7	FSOURCE_D	E7	GNDK	F7	VCCK
G7	GPIO7	H7	GPIO9	J7	GPIO10
A8	VIO18	B8	TESTMODE	C8	AVDD18_TCXO
D8	RTC_XIN	G8	VSRAM	H8	GPIO8
J8	GPIO11	K8	GPIO6	A9	VCORE
B9	AVSS_AUXADC	C9	EXT_PWR_EN	D9	RTC_XOUT
E9	AVDD18_USB	F9	VUSB	G9	GPIO0
H9	GPIO1	J9	GPIO4	K9	GPIO5
A10	AVDD18_AUXADC	B10	TSENSE_IN	C10	VRTC
E10	USB20_DM	F10	USB20_DP	H10	GPIO2
J10	GPIO3	K10	DVDD_MC0		

6.1.2 AG3335A pin functions

Table 6.1-2. Acronym for pin types and I/O structure

Name	Abbreviation	Description
Pin Type	AI	Analog input
	AO	Analog output
	AIO	Analog bi-direction
	DI	Digital input
	DO	Digital output
	DIO	Digital bi-direction
	P	Power
	G	Ground

Name	Abbreviation	Description
IO Structure	TYPE0	Pull-up/down 1.98V tolerance Analog input/output
	TYPE1	Pull-up/down 3.08V tolerance

Table 6.1-3. AG3335A series pin function description and power domain (group1)

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
RTC						
D9	RTC_XOUT	AIO	-	Input pin for 32.768kHz crystal	-	VRTC
D8	RTC_XIN	AIO	-	Input pin for 32.768kHz crystal	-	VRTC
A5	RTC_EINT	DI	-	RTC external interrupt	-	AVDD18_VR TC_IN
RF GPS Interface						
A2	RF_IN_L1	AI	-	RF input of L1 channel	-	AVDD07_L1 _L5
A3	RF_IN_L5	AI	-	RF input of L5 channel	-	AVDD07_L1 _L5
B2	AVDD07_L1_L5	P	-	RF block power supply from PMU	-	-
C2	AVDD07_ANA	P	-	Analog block power supply from PMU	-	-
B1	AVDD07_LO	P	-	SX block power supply from PMU	-	-
D3	AVDD07_DCXO	P	-	DCXO power supply from PMU	-	-
C1	OSC_XIN	AIO	-	Input from XTAL	-	AVDD07_DC XO
D2	OSC_XOUT	AIO	-	Drive for XTAL	-	- AVDD07_DC XO
C5	FREF0	AIO	-	External reference clk0 buffer output	-	VIO
E4	FREF1	AIO	-	External reference clk1 buffer output	-	VIO
C3	AVSS_RF_ANA	G	-	RF analog ground	-	-
D4	AVSS_RF_DCXO	G	-	RF DCXO ground	-	-
A1, B3, B4	AVSS_RF_L1_L5	G	-	RF L1 and L5 ground	-	-
PMU						
C4	CHIP_EN	AI	-	Chip enable	-	AVDD18_VR TC_IN

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
B5	VRF	P	-	Output of LDO of VRF	-	VDD_VRF_IN
A6	AVDD18_IN	P	-	Input power of bandgap	-	-
B6	AVDD18_VRTC_IN	P	-	Input power of LDO of VRTC	-	AVDD18_VRTC_IN
C6	AVSS18_VSYS	G	-	VSYS analog ground	-	-
B7	VDD_VRF_IN	P	-	Input power of LDO of VRF	-	-
C7	VDD_SYS_IN	P	-	Input power of LDO of VCORE and VSRAM	-	-
A8	VIO18	P	-	Output of power switch of VIO18	-	AVDD18_IN
B8	TESTMODE	AI	-	Reserved for test mode. Tied to GND in normal mode	-	AVDD18_VRTC_IN
C8	AVDD18_TCXO	P	-	Output of power switch of TCXO	-	AVDD18_IN
A9	VCORE	P	-	Output of LDO of VCORE	-	VDD_SYS_IN
A10	AVDD18_AUXADC	P	-	Input power for AUXADC	-	-
B9	AVSS_AUXADC	G	-	Ground for AUXADC	-	-
B10	TSENSE_IN	AI	-	Thermal sensor input	-	AVDD18_AUXADC
C9	EXT_PWR_EN	AO	-	External power enable	-	AVDD18_VRTC_IN
C10	VRTC	P	-	Output of LDO of VRTC	-	AVDD18_VRTC_IN
E9	AVDD18_USB	P	-	Input power for power switch of USB 1.8v	-	-
F9	VUSB	P	-	Input power for USB	-	AVDD18_USB
Digital Power Supply						
D7	FSOURCE_D	P	-	eFuse power source	-	-
K1	DVDD18_IO	P	-	Power input of GPIO group 0	-	-
K10	DVDD_MC0	P	-	Power input of MSDC GPIO group	-	-
F7	VCKK	P	-	VCore power	-	-
G8	VSRAM	P	-	SRAM power	-	-
E7, F5, H3	GNDK	G	-	Digital ground	-	-
USB 2.0						
E10	USB20_DM	AIO	-	USB signal DM	-	VUSB
F10	USB20_DP	AIO	-	USB signal DP	-	VUSB
General Programmable I/O						
G9	GPIO0	DIO	TYPE 1	General purpose input/output, Pin 0	MSDC (0) SPI Slave (0) UART (0)	DVDD18_IO/ DVDD_MC0

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
					PWM (0) UART (2) CM4 JTAG	
H9	GPIO1	DIO	TYPE 1	General purpose input/output, Pin 1	MSDC (0) SPI Slave (0) UART (0) L1_LNA_EN UART (2) CM4 JTAG	DVDD18_IO/ DVDD_MC0
H10	GPIO2	DIO	TYPE 1	General purpose input/output, Pin 2	MSDC (0) SPI Slave (0) UART (0) I2C Slave (0) UART (1) CM4 JTAG	DVDD18_IO/ DVDD_MC0
J10	GPIO3	DIO	TYPE 1	General purpose input/output, Pin 3	MSDC (0) SPI Slave (0) UART (0) I2C Slave (0) UART (1) SPI Master (0) CM4 JTAG	DVDD18_IO/ DVDD_MC0
J9	GPIO4	DIO	TYPE 1	General purpose input/output, Pin 4	MSDC (0) SPI Slave (0) PWM (1) WDTRSTB CM4 JTAG	DVDD18_IO/ DVDD_MC0
K9	GPIO5	DIO	TYPE 1	General purpose input/output, Pin 5	MSDC (0) SPI Slave (0) SRCLKENA_WIFI UART (2) I2C Master (0) L5_LNA_EN	DVDD18_IO/ DVDD_MC0
K8	GPIO6	DIO	TYPE 1	General purpose input/output, Pin 6	MSDC (0) ESC SPI Master (0) UART (1) UART (2) I2C Master (0)	DVDD18_IO/ DVDD_MC0
G7	GPIO7	DIO	TYPE 1	General purpose input/output, Pin 7	I2C Slave (0) ESC SPI Master (0) UART (1) UART (0) SPI Slave (0) MSDC (0)	DVDD18_IO/ DVDD_MC0

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
H8	GPIO8	DIO	TYPE 1	General purpose input/output, Pin 8	I2C Slave (0) ESC SPI Master (0) UART (1) UART (0) SPI Slave (0) MSDC (0)	DVDD18_IO/ DVDD_MC0
H7	GPIO9	DIO	TYPE 1	General purpose input/output, Pin 9	I2C Master (1) ESC SPI Master (0) UART (1) SRCLKENA_WIFI I2C Slave (0) MSDC (0)	DVDD18_IO/ DVDD_MC0
J7	GPIO10	DIO	TYPE 1	General purpose input/output, Pin 10	I2C Master (1) ESC SPI Master (0) PPS (0) SRCLKENA_NBIOT I2C Slave (0) MSDC (0)	DVDD18_IO/ DVDD_MC0
J8	GPIO11	DIO	TYPE 1	General purpose input/output, Pin 11	ESC SPI Master (0) PPS (1) RESETB_OUT MSDC (0)	DVDD18_IO/ DVDD_MC0
G6	GPIO12	DIO	TYPE 0	General purpose input/output, Pin 12	SPI Slave (0) SPI Master (0) SPI Master (1) CLKO (1)	DVDD18_IO
H6	GPIO13	DIO	TYPE 0	General purpose input/output, Pin 13	SPI Slave (0) SPI Master (0) SPI Master (1) CLKO (0) DCXO_REF_10M RESETB_OUT DSP L5 JTAG	DVDD18_IO
J6	GPIO14	DIO	TYPE 0	General purpose input/output, Pin 14	SPI Slave (0) SPI Master (0) SPI Master (1) I2C Master (0) L1_LNA_EN NB_L1_BLANKING DSP L5 JTAG	DVDD18_IO
K6	GPIO15	DIO	TYPE 0	General purpose input/output, Pin 15	SPI Slave (0) SPI Master (0) SPI Master (1) I2C Master (0)	DVDD18_IO

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
					L5_LNA_EN NB_L5_BLANKING DSP L5 JTAG	
G5	GPIO16	DIO	TYPE 0	General purpose input/output, Pin 16	SPI Slave (0) SPI Master (0) SPI Master (1) I2C Master (1) UART (1) UART (2) DSP L5 JTAG	DVDD18_IO
K5	GPIO17	DIO	TYPE 0	General purpose input/output, Pin 17	SPI Slave (0) SPI Master (0) SPI Master (1) I2C Master (1) UART (1) UART (2) DSP L5 JTAG	DVDD18_IO
J5	GPIO18	DIO	TYPE 0	General purpose input/output, Pin 18	SPI Master (1) ESC UART (2) I2C Slave (0) WDTRSTB SPI Master (0) NB_L5_BLANKING	DVDD18_IO
H4	GPIO19	DIO	TYPE 0	General purpose input/output, Pin 19	SPI Master (1) ESC UART (2) I2C Slave (0) DSP L1 JTAG WDTRSTB NB_L1_BLANKING	DVDD18_IO
J4	GPIO20	DIO	TYPE 0	General purpose input/output, Pin 20	SPI Master (1) ESC DSP L1 JTAG RESETB_OUT	DVDD18_IO
J3	GPIO21	DIO	TYPE 0	General purpose input/output, Pin 21	SPI Master (1) ESC DSP L1 JTAG	DVDD18_IO
K3	GPIO22	DIO	TYPE 0	General purpose input/output, Pin 22	SPI Master (1) ESC UART (0) UART (1) DSP L1 JTAG NB_FRAME_SYNC	DVDD18_IO
K2	GPIO23	DIO	TYPE 0	General purpose input/output, Pin 23	SPI Master (1) ESC UART (0) UART (1)	DVDD18_IO

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
					DSP L1 JTAG SPI Master (0)	
J2	GPIO24	DIO	TYPE 0	General purpose input/output, Pin 24	PWM (1) VCORE_PRE_ON_RSTB SRCLKENA_PMU DCXO_REF_10M NB_FRAME_SYNC	DVDD18_IO
F4	GPIO25	DIO	TYPE 0	General purpose input/output, Pin 25	PWM (2) DCXO_REF_10M CM4 JTAG WDTRSTB CLKO (3) DSP L5 JTAG	DVDD18_IO
E2	GPIO26	DIO	TYPE 0	General purpose input/output, Pin 26	NB_FRAME_SYNC AUXADC (1) CM4 JTAG UART (0) NB_L5_BLANKING SPI Master (0) DSP L5 JTAG	DVDD18_IO
F2	GPIO27	DIO	TYPE 0	General purpose input/output, Pin 27	L1_LNA_EN AUXADC (3) CM4 JTAG I2C Slave (0) I2C Master (0) SPI Master (1) DSP L5 JTAG	DVDD18_IO
F1	GPIO28	DIO	TYPE 0	General purpose input/output, Pin 28	NB_L1_BLANKING AUXADC (2) CM4 JTAG I2C Slave (0) I2C Master (0) SPI Master (1) DSP L5 JTAG	DVDD18_IO
E1	GPIO29	DIO	TYPE 0	General purpose input/output, Pin 29	L5_LNA_EN AUXADC (0) CM4 JTAG UART (0) PWM (0) SPI Master (1) DSP L5 JTAG	DVDD18_IO
G4	GPIO30	DIO	TYPE 0	General purpose input/output, Pin 30	SRCLKENA_WIFI PWM (0) UART (1) UART (0) I2C Master (0) DSP L1 JTAG	DVDD18_IO

Pin Number	Pin Name	Pin Type	IO Structure	Pin Description	Alternate Pin Functions	Power domain
AG3335A						
J1	GPIO31	DIO	TYPE 0	General purpose input/output, Pin 31	SRCLKENA_NBIOT PWM (1) UART (1) UART (0) I2C Master (0) DSP L1 JTAG	DVDD18_IO
H1	GPIO32	DIO	TYPE 0	General purpose input/output, Pin 32	L1_LNA_EN UART (2) UART (1) CLKO (2) I2C Slave (0) DSP L1 JTAG	DVDD18_IO
H2	GPIO33	DIO	TYPE 0	General purpose input/output, Pin 33	L5_LNA_EN UART (2) UART (1) CLKO (3) I2C Slave (0) DSP L1 JTAG	DVDD18_IO
G3	GPIO34	DIO	TYPE 0	General purpose input/output, Pin 34	PWM (2) UART (2) CLKO (1) SPI Master (0) DSP L1 JTAG	DVDD18_IO
G2	GPIO35	DIO	TYPE 0	General purpose input/output, Pin 35	SRCLKENA_WIFI UART (2) CLKO (0) I2C Slave (0) CM4 JTAG	DVDD18_IO
F3	GPIO36	DIO	TYPE 0	General purpose input/output, Pin 36	SRCLKENA_NBIOT RESETB_OUT DCXO_REF_10M SPI Master (0) I2C Slave (0) CM4 JTAG	DVDD18_IO

6.1.3 AG3335A pin multiplexing

AG3335A platform offers 37 GPIO pins. By setting up the control registers, the MCU software can control the direction, the output value, and read the input values on the pins. The GPIOs and GPOs are multiplexed with other functions to reduce the pin count. To facilitate application use, the software can configure which clock to send outside the chip. There are four clock-out ports embedded in GPIO pins and each clock-out can be programmed to output an appropriate clock source. In addition, when two GPIOs function for the same peripheral IP, the smaller GPIO serial number has higher priority over the bigger one.

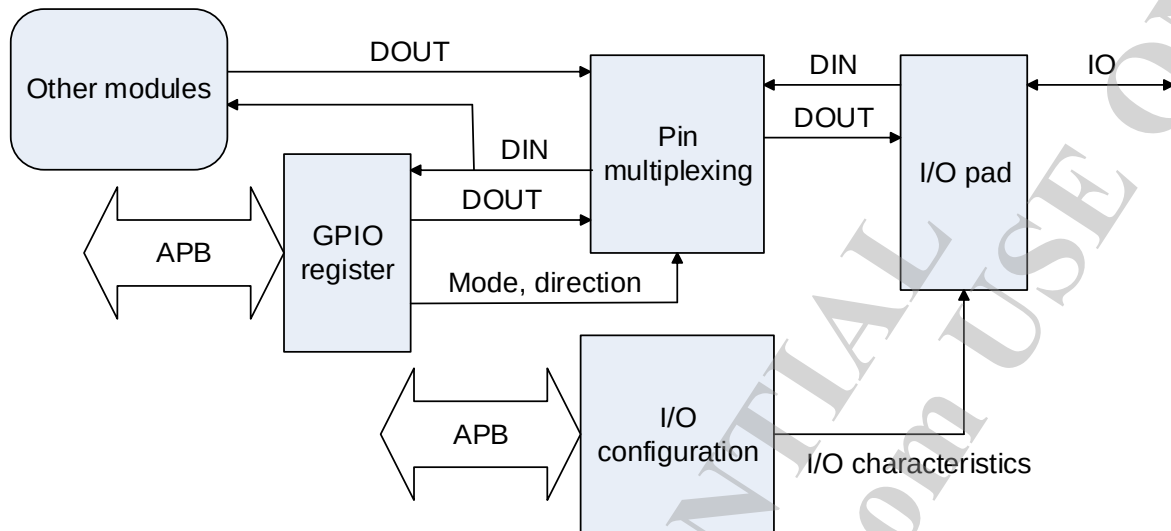


Figure 6.1-2. GPIO block diagram

AG3335A has rich peripheral functions and the peripheral signals are shown in Table 6.1-4.

The MSDC, SPI Master, and SPI Slave can support signal group allocating on different pins.

Table 6.1-4. Peripheral functions and signals

Alternate Function	Signal List	Alternate Function	Signal List
MSDC (0)	MSDC0_CLK MSDC0_CMD MSDC0_DAT0 MSDC0_DAT1 MSDC0_DAT2 MSDC0_DAT3 MSDC0_RST	SPI Master (0)	SPI_MST0_SCK SPI_MST0_CS SPI_MST0_MISO SPI_MST0_MOSI SPI_MST0_SIO2 SPI_MST0_SIO3
ESC	ESC_SIO2 ESC_SIO3 ESC_MISO ESC_MOSI ESC_SCK ESC_CS	SPI Master (1)	SPI_MST1_SCK SPI_MST1_CS SPI_MST1_MISO SPI_MST1_MOSI SPI_MST1_SIO2 SPI_MST1_SIO3
SPI Slave (0)	SPI_SLV0_SCK SPI_SLV0_CS SPI_SLV0_MISO SPI_SLV0_MOSI SPI_SLV0_SIO2 SPI_SLV0_SIO3	PWM (0)	PWM0
UART (0)	UART0_RXD	PWM (1)	PWM1
		PWM (2)	PWM2
		AUXADC	AUXADC0 AUXADC1 AUXADC2 AUXADC3
		CM4 JTAG	AP_JTDI

	UART0_TXD UART0_RTS UART0_CTS
UART (1)	UART1_RXD UART1_TXD UART1_RTS UART1_CTS
UART (2)	UART2_RXD UART2_TXD UART2_RTS UART2_CTS
I2C (0)	I2C0_SCL I2C0_SDA
I2C (1)	I2C1_SCL I2C1_SDA
I2C (2)	I2C2_SCL I2C2_SDA
I2C_SLV	I2C_SLV0_SCL I2C_SLV0_SDA

	AP_JRSTB AP_JTDO AP_JTMS AP_JTCK
DSP L1 JTAG	DSP_L1_JMS DSP_L1_JCK DSP_L1_JDI DSP_L1_JINTP DSP_L1_JDO
DSP L5 JTAG	DSP_L5_JMS DSP_L5_JCK DSP_L5_JDI DSP_L5_JINTP DSP_L5_JDO

Table 6.1-5. PinMux description

Pin Name	Mode 0	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7	Mode 8
GPIO0	GPIO0	MSDC0_CMD	SPI_SLV0_MOSI	UART0_RXD	PWM0	UART2_RTS	MSDC0_CLK1	AP_JTMS	EINT0
GPIO1	GPIO1	MSDC0_CLK0	SPI_SLV0_SIO2	UART0_TXD	L1_LNA_EN	UART2_CTS	MSDC0_CLK2	AP_JTCK	EINT1
GPIO2	GPIO2	MSDC0_DAT1	SPI_SLV0_SCK	UART0_RTS	I2C_SLV0_SCL	UART1_RXD	MSDC0_CLK3	AP_JRSTB	EINT2
GPIO3	GPIO3	MSDC0_DAT0	SPI_SLV0_CS	UART0_CTS	I2C_SLV0_SDA	UART1_TXD	SPI_MST0_CS1	AP_JTDI	EINT3
GPIO4	GPIO4	MSDC0_RST	SPI_SLV0_MISO		PWM1		WDTRSTB	AP_JTDO	EINT4
GPIO5	GPIO5	MSDC0_DAT3	SPI_SLV0_SIO3	SPI_SLV0_MISO		UART2_TXD	I2C_MST0_SDA	L5_LNA_EN	EINT5
GPIO6	GPIO6	MSDC0_DAT2	ESC_SIO2	SPI_MST0_SIO3	UART1_RXD	UART2_RXD	I2C_MST0_SCL		EINT6
GPIO7	GPIO7	I2C_SLV0_SCL	ESC_SIO3	SPI_MST0_SIO2	UART1_RTS	UART0_TXD	SPI_SLV0_MISO	MSDC0_DAT1	EINT7
GPIO8	GPIO8	I2C_SLV0_SDA	ESC_MISO	SPI_MST0_MISO	UART1_CTS	UART0_RXD	SPI_SLV0_MOSI	MSDC0_DAT0	EINT8
GPIO9	GPIO9	I2C_MST1_SCL	ESC_MOSI	SPI_MST0_CS0	UART1_TXD		I2C_SLV0_SCL	MSDC0_CMD	EINT9
GPIO10	GPIO10	I2C_MST1_SDA	ESC_SCK	SPI_MST0_SCK	PPS0 (±2ns accuracy)		I2C_SLV0_SDA	MSDC0_CLK0	EINT10
GPIO11	GPIO11		ESC_CS	SPI_MST0_MOSI	PPS1 (±2ns accuracy)		RESETB_OUT	MSDC0_RST	EINT11
GPIO12	GPIO12	SPI_SLV0_MISO	SPI_MST0_MOSI		CLKO1		SPI_MST1_MOSI		EINT12
GPIO13	GPIO13	SPI_SLV0_MOSI	SPI_MST0_MISO	DCXO_REF_10M	CLKO0	RESETB_OUT	SPI_MST1_MISO	DSP_L5_JMS	EINT13
GPIO14	GPIO14	SPI_SLV0_SIO2	SPI_MST0_SIO2	L1_LNA_EN	I2C_MST0_SCL	NB_L1_BLANKING	SPI_MST1_SIO2	DSP_L5_JCK	EINT14
GPIO15	GPIO15	SPI_SLV0_SIO3	SPI_MST0_SIO3	L5_LNA_EN	I2C_MST0_SDA	NB_L5_BLANKING	SPI_MST1_SIO3	DSP_L5_JDI	EINT15
GPIO16	GPIO16	SPI_SLV0_SCK	SPI_MST0_SCK	UART2_RTS	UART1_RXD	I2C_MST1_SCL	SPI_MST1_SCK	DSP_L5_JINTP	EINT16
GPIO17	GPIO17	SPI_SLV0_CS	SPI_MST0_CS0	UART2_CTS	UART1_TXD	I2C_MST1_SDA	SPI_MST1_CS0	DSP_L5_JDO	EINT17
GPIO18	GPIO18	SPI_MST1_MISO	ESC_MISO	UART2_TXD	I2C_SLV0_SCL	WDTRSTB	SPI_MST0_CS2	NB_L5_BLANKING	EINT18
GPIO19	GPIO19	SPI_MST1_MOSI	ESC_SCK	UART2_RXD	I2C_SLV0_SDA	DSP_L1_JDI	WDTRSTB	NB_L1_BLANKING	EINT19
GPIO20	GPIO20	SPI_MST1_SIO2	ESC_CS			DSP_L1_JINTP	RESETB_OUT		EINT20
GPIO21	GPIO21	SPI_MST1_SIO3	ESC_MOSI			DSP_L1_JDO			EINT21
GPIO22	GPIO22	SPI_MST1_SCK	ESC_SIO2	UART0_RXD	UART1_CTS	DSP_L1_JCK	NB_FRAME_SYNC		EINT22

Pin Name	Mode 0	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7	Mode 8
GPIO23	GPIO23	SPI_MST1_CS0	ESC_SIO3	UART0_TXD	UART1_RTS	DSP_L1_JMS	SPI_MST0_CS1		EINT23
GPIO24	GPIO24	PWM1	VCORE_PRE_ON_RSTB	SRCLKENA_PMU	DCXO_REF_10M	NB_FRAME_SYNC			EINT24
GPIO25	GPIO25	PWM2	DCXO_REF_10M	AP_JTDO	WDTRSTB	CLKO3		DSP_L5_JDO	EINT25
GPIO26	GPIO26	NB_FRAME_SYNC	AUXADC1	AP_JTCK	UART0_TXD	NB_L5_BLANKING	SPI_MST0_CS3	DSP_L5_JCK	EINT26
GPIO27	GPIO27	L1_LNA_EN	AUXADC3	AP_JTDI	I2C_SLV0_SCL	I2C_MST0_SCL	SPI_MST1_CS1	DSP_L5_JDI	EINT27
GPIO28	GPIO28	NB_L1_BLANKING	AUXADC2	AP_JRSTB	I2C_SLV0_SDA	I2C_MST0_SDA	SPI_MST1_CS2	DSP_L5_JINTP	EINT28
GPIO29	GPIO29	L5_LNA_EN	AUXADC0	AP_JTMS	UART0_RXD	PWM0	SPI_MST1_CS3	DSP_L5_JMS	EINT29
GPIO30	GPIO30	SRCLKENA_WiFi	PWM0	UART1_RTS	UART0_TXD		I2C_MST0_SCL	DSP_L1_JINTP	EINT30
GPIO31	GPIO31	SRCLKENA_NBIOT	PWM1	UART1_CTS	UART0_RXD		I2C_MST0_SDA	DSP_L1_JCK	EINT31
GPIO32	GPIO32	L1_LNA_EN	UART2_CTS	UART1_RXD	CLKO2		I2C_SLV0_SCL	DSP_L1_JMS	EINT32
GPIO33	GPIO33	L5_LNA_EN	UART2_RTS	UART1_TXD	CLKO3		I2C_SLV0_SDA	DSP_L1_JDO	EINT33
GPIO34	GPIO34	PWM2	UART2_RXD		CLKO1		SPI_MST0_CS3	DSP_L1_JDI	EINT34
GPIO35	GPIO35	SRCLKENA_WiFi	UART2_TXD		CLKO0		I2C_SLV0_SCL	AP_JTMS	EINT35
GPIO36	GPIO36	SRCLKENA_NBIOT	RESETB_OUT	DCXO_REF_10M	SPI_MST0_CS2		I2C_SLV0_SDA	AP_JTCK	EINT36

Table 6.1-6. IO default state

Pin Name	Default value for IO PinMUX mode	Default IO PinMUX mode	Default IO direction	Default IO pull direction	Note
GPIO0	0	GPIO0	input	Pull down	
GPIO1	0	GPIO1	input	Pull down	
GPIO2	0	GPIO2	input	Pull down	
GPIO3	0	GPIO3	input	Pull down	
GPIO4	0	GPIO4	input	Pull down	
GPIO5	0	GPIO5	input	Pull down	

Pin Name	Default value for IO PinMUX mode	Default IO PinMUX mode	Default IO direction	Default IO pull direction	Note
GPIO6	0	GPIO6	input	Pull down	
GPIO7	5	UART0_TXD	Output	Drive high	
GPIO8	5	UART0_RXD	input	Pull up	
GPIO9	0	GPIO9	input	Pull down	
GPIO10	0	GPIO10	input	Pull down	
GPIO11	0	GPIO11	input	Pull down	
GPIO12	0	GPIO12	input	Pull down	
GPIO13	5	RESETB_OUT	Output	Drive high	
GPIO14	0	GPIO14	input	Pull down	
GPIO15	0	GPIO15	input	Pull down	
GPIO16	0	GPIO16	input	Pull down	
GPIO17	0	GPIO17	input	Pull down	
GPIO18	0	GPIO18	input	Pull down	
GPIO19	0	GPIO19	input	Pull down	
GPIO20	0	GPIO20	input	Pull down	
GPIO21	0	GPIO21	input	Pull down	
GPIO22	0	GPIO22	input	Pull down	
GPIO23	0	GPIO23	input	Pull down	
GPIO24	0	GPIO24	input	Pull down	
GPIO25	0	GPIO25	input	Pull down	
GPIO26	3	UART0_TXD	input	Pull up	
GPIO27	0	GPIO27	input	Pull down	
GPIO28	0	GPIO28	input	Pull down	
GPIO29	3	UART0_RXD	input	Up	
GPIO30	0	GPIO30	input	Pull down	

Pin Name	Default value for IO PinMUX mode	Default IO PinMUX mode	Default IO direction	Default IO pull direction	Note
GPIO31	0	GPIO31	input	Pull down	
GPIO32	0	GPIO32	input	Pull down	
GPIO33	0	GPIO33	input	Pull down	
GPIO34	0	GPIO34	input	Pull down	
GPIO35	0	GPIO35	input	Pull down	
GPIO36	0	GPIO36	input	Pull down	

7 Electrical Characteristics

7.1 Absolute maximum ratings

7.1.1 AG3335A

Table 7.1-1. Absolute maximum ratings for power supply

Symbol or pin name	Description	Min.	Max.	Unit
AVDD07_L1_L5	RF power input	0	0.77	V
AVDD07_ANA	SX power input	0	0.77	V
AVDD07_LO	DCXO power input	0	0.77	V
AVDD07_DCXO	Power supply of RF	0	0.77	V
VRF	LDO_VRF VOUT	0	0.77	V
AVDD18_IN	Power supply of BG	0	1.98	V
AVDD18_VRTC_IN	Power supply of LDO_VRTC	0	1.98	V
VDD_VRF_IN	Power supply of LDO_VRF	0	1.98	V
VDD_SYS_IN	Power supply of LDO_VCORE, LDO_VSRAM	0	1.98	V
VIO18	Power supply of IO	0	1.98	V
AVDD18_TCXO	Power supply of TCXO	0	1.98	V
VCORE	LDO_VCORE VOUT and Core power input	0	0.88	V
AVDD18_AUXADC	AUXADC Power input	0	1.98	V
AVDD18_USB	Power supply of PSW_USB18	0	1.98	V
VUSB	Power supply of USB	0	3.37	V
VCCK	Core power input	0	0.88	V
VSRAM	LDO_VSRAM VOUT and SRAM power input	0	1.05	V
FSOURCE_D	Power supply of eFuse blow resource	0	1.89	V

Table 7.1-2. Absolute maximum ratings for I/O power supply

Symbol or pin name	Description	Min.	Typ.1	Typ.2	Max.	Unit
DVDD18_IO	Power supply for GPIO group 0	1.62	1.8	-	1.98	V
DVDD_MC0	Power input of MSDC GPIO group	1.62	1.8	2.8	3.08	V

Table 7.1-3. Absolute maximum ratings for voltage input

Symbol or pin name	Description	Min.	Max.	Unit
VIN0	Digital input voltage for IO Type 0	-0.3	1.98	V
VIN1	Digital input voltage for IO Type 1	-0.3	3.63	V

Table 7.1-4. Absolute maximum ratings for storage temperature

Symbol or pin name	Description	Min.	Max.	Unit
Tstg	Storage temperature	-55	125	°C

7.1.1.1 Input or output port characteristics

Table 7.1.1-1. Electrical characteristics

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
DIIH0	Digital high input current for IO Type 0	PU/PD disabled DVDIO = 1.8V, DVDIO*0.65 < VIN0 < DVDIO	-16	-	16	μA
		PU enabled DVDIO = 1.8V DVDIO*0.65 < VIN0 < DVDIO	-30	-	16	μA
		PD enabled DVDIO = 1.8V DVDIO*0.65 < VIN0 < DVDIO	-16	-	45	μA
DIILO	Digital low input current for IO Type 0	PU/PD disabled DVDIO = 1.8V, 0V < VIN0 < DVDIO*0.35	-16	-	16	μA
		PU enabled, DVDIO = 1.8V 0 < VIN0 < DVDIO*0.35	-45	-	16	μA
		PD enabled, DVDIO = 1.8V 0 < VIN0 < DVDIO*0.35	-16	-	30	μA
DIOH0	Digital high output current for IO Type 0	DVOH = 1.35V DVDIO = 1.8V Max. driving mode	16	-	-	mA
DIOLO	Digital low output current for IO Type 0	DVOL = 0.45V DVDIO = 1.8V Max. driving mode	16	-	-	mA
DRPU0	Digital I/O pull-up resistance for IO Type 0	DVDIO = 1.8V VIN0 = 0V	40	75	190	kΩ
DRPD0	Digital I/O pull-down resistance for IO Type 0	DVDIO = 1.8V VIN0 = 1.8V	40	75	190	kΩ
DVOH0	Digital output high voltage for IO Type 0	DVDIO = 1.8V	0.75*D VDD	-	-	V

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
DVOL0	Digital output low voltage for IO Type 0	DVDIO = 1.8V	-	-	0.25*D VDIO	V
VIH0	Digital input high voltage for IO Type 0	DVDIO = 1.8V	0.65*D VDIO		DVDIO +0.3	
VIL0	Digital input low voltage for IO Type 0	DVDIO = 1.8V	-0.3		0.35*D VDIO	
DIIH1	Digital high input current for IO Type 1	PU/PD disabled DVDIO = 2.8/1.8V, DVDIO*0.65 < VIN1 < DVDIO	-16	-	16	μA
		PU enabled, RSEL1 DVDIO = 2.8/1.8V DVDIO*0.65 < VIN1 < DVDIO	-170	-	16	μA
		PU enabled, RSEL2 DVDIO = 2.8/1.8V DVDIO*0.65 < VIN1 < DVDIO	-35	-	16	μA
		PD enabled, RSEL1 DVDIO = 2.8/1.8V DVDIO*0.65 < VIN1 < DVDIO	-16	-	500	μA
		PD enabled, RSEL2 DVDIO = 2.8/1.8V DVDIO*0.65 < VIN1 < DVDIO	-16	-	80	μA
DIIL1	Digital low input current for IO Type 1	PU/PD disabled, DVDIO = 2.8/1.8V, 0V < VIN1 < DVDIO*0.35	-16	-	16	μA
		PU enabled, RSEL1 DVDIO = 2.8/1.8V 0 < VIN1 < DVDIO*0.35	-500	-	16	μA
		PU enabled, RSEL2 DVDIO = 2.8/1.8V 0 < VIN1 < DVDIO*0.35	-80	-	16	μA
		PD enabled, RSEL1 DVDIO = 2.8/1.8V 0 < VIN1 < DVDIO*0.35	-16	-	170	μA
		PD enabled, RSEL2 DVDIO = 2.8/1.8V 0 < VIN1 < DVDIO*0.35	-16	-	35	μA
DIOH1	Digital high output current for IO Type 1	DVOH = 2.1V DVDIO = 2.8V	12	-	-	mA

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		Max. driving mode				
		DVOH = 1.35V DVDIO = 1.8V Max. driving mode	10	-	-	mA
DIOL1	Digital low output current for IO Type 1	DVOL = 0.35V DVDIO = 2.8V Max. driving mode	7	-	-	mA
		DVOL = 0.45V DVDIO = 1.8V Max. driving mode	10	-	-	mA
DRPU1	Digital I/O pull-up resistance for IO Type 1 RSEL1 : (GPIO_R1, GPIO_R0) = (0,1) RSEL2 : (GPIO_R1, GPIO_R0) = (1, 0)	DVDIO = 2.8V VIN1 = 0V, RSEL1	5	7.5	10	kΩ
		DVDIO = 2.8V VIN1 = 0V, RSEL2	10	50	100	kΩ
		DVDIO = 1.8V VIN1 = 0V, RSEL1	5	7.5	10	kΩ
		DVDIO = 1.8V VIN1 = 0V, RSEL2	10	50	100	kΩ
DRPD1	Digital I/O pull-down resistance for IO Type 1 RSEL1 : (GPIO_R1, GPIO_R0) = (0,1) RSEL2 : (GPIO_R1, GPIO_R0) = (1, 0)	DVDIO = 2.8V VIN1 = 2.8V, RSEL1	5	7.5	10	kΩ
		DVDIO = 2.8V VIN1 = 2.8V, RSEL2	10	50	100	kΩ
		DVDIO = 1.8V VIN1 = 1.8V, RSEL1	5	7.5	10	kΩ
		DVDIO = 1.8V VIN1 = 1.8V, RSEL2	10	50	100	kΩ
DVOH1	Digital output high voltage for IO Type 1	DVDIO = 2.8V	0.75*D VDIO	-	-	V
		DVDIO = 1.8V	0.75*D VDD	-	-	V
DVOL1	Digital output low voltage for IO Type 1	DVDIO = 2.8V	-	-	0.125* DVDIO	V
		DVDIO = 1.8V	-	-	0.25*D VDIO	V
VIH1	Digital input high voltage for IO Type 1	DVDIO = 2.8V	0.625* DVDIO		DVDIO +0.3	
		DVDIO = 1.8V	0.65*D VDIO		DVDIO +0.3	
VIL1	Digital input low voltage for IO Type 1	DVDIO = 2.8V	-0.3		0.25*D VDIO	
		DVDIO = 1.8V	-0.3		0.35*D VDIO	

7.2 Electrostatic discharge (ESD) ratings

Table 7.2-1. ESD electrical characteristic of AG3335A

ESD mode		Pin name	Min.	Max.	Unit
Human body model (HBM)	JESD22-A114-F	All pins	-1000	1000	V
Charged device model (CDM)	JESD22-C101-D	All pins exclude RF pins	-500	500	V
		RF pins	-250	250	V

7.3 Operating conditions

7.3.1 AG3335A

7.3.1.1 General operating conditions

Table 7.3-1. AG3335A general operating conditions

Item	Description	Condition	Min.	Typ.	Max.	Unit
F _{CPU}	Internal Cortex-M4 clock (including its cache and TCM)	V _{CORE} = 0.65V	0	-	192	MHz
		V _{CORE} = 0.7V	0	-	384	MHz
		V _{CORE} = 0.83V	0	-	530	MHz
F _{BUS}	Internal system memory and system bus clock. Synchronous with F _{CPU} .	V _{CORE} = 0.65V	0	-	96	MHz
		V _{CORE} = 0.7V	0	-	192	MHz
		V _{CORE} = 0.83V	0	-	265	MHz
F _{EMI}	Internal EMI clock. Asynchronous with F _{CPU} .	V _{CORE} = 0.65V	0	-	192	MHz
		V _{CORE} = 0.7V	0	-	192	MHz
		V _{CORE} = 0.83V	0	-	192	MHz
F _{SFC}	Internal SFC clock. Asynchronous with F _{CPU} .	V _{CORE} = 0.65V	0	-	104	MHz
		V _{CORE} = 0.7V	0	-	104	MHz
		V _{CORE} = 0.83V	0	-	104	MHz

Table 7.3-2. Recommended operating conditions for power supply

Symbol or pin name	Description	Min.	Typ.	Max.	Unit
AVDD07_L1_L5	RF Power net from PMU_TOP	0	0.7	0.77	V
AVDD07_ANA	Analog Power net from PMU_TOP	0	0.7	0.77	V
AVDD07_LO	SX Power net from PMU_TOP	0	0.7	0.77	V
AVDD07_DCXO	DCXO Power net from PMU_TOP	0	0.7	0.77	V
VRF	LDO_VRF VOUT	0	0.7	0.77	V
AVDD18_IN	Power supply of BG	1.72	1.8	1.98	V
AVDD18_VRTC_IN	Power supply of LDO_VRTC	1.62	1.8	1.98	V
VDD_VRF_IN	Power supply of LDO_VRF	0.7	1.8	1.98	V
VDD_SYS_IN	Power supply of LDO_VCORE, LDO_VSRAM	1.15	1.8	1.98	V
VIO18	Power supply of IO	1.62	1.8	1.98	V
AVDD18_TCXO	PSW_ANA VOUT	1.70	1.8	1.98	V
AVDD18_AUXADC	AUXADC power	1.68	1.8	1.98	V
VRTC	LDO_RTC VOUT	0.63	0.75	0.88	V
AVDD18_USB	Power supply of PSW_USB18	1.73	1.8	1.98	V
VUSB	Power supply of USB	2.92	3.07	3.22	V

Symbol or pin name	Description	Min.	Typ.	Max.	Unit
VCKK VCORE	Digital Core power (0.65V scenario)	0.62	0.65	0.71	V
	Digital Core power (0.7V scenario)	0.67	0.7	0.77	V
	Digital Core power (0.83V scenario)	0.79	0.83	0.88	V
VSRAM	LDO_VSRAM VOUT (0.65V scenario)	0.8	0.85	0.935	V
	LDO_VSRAM VOUT (0.7V scenario)	0.8	0.85	0.935	V
	LDO_VSRAM VOUT (0.83V scenario)	0.88	0.93	1.05	V
FSOURCE_D	Power supply of eFuse blow resource	1.71	1.8	1.89	V

Table 7.3-3. Recommended operating conditions for voltage input

Symbol or pin name	Description	Min.	Typ.	Max.	Unit
VIN0	Digital input voltage for IO Type 0	0	-	DVDIO	V
VIN1	Digital input voltage for IO Type 1	0	-	DVDIO	V
VIN2	Digital input voltage for IO Type 2	0	-	DVDIO	V

Table 7.3-4. Recommended operating conditions for operating temperature

Symbol or pin name	Description	Min.	Typ.	Max.	Unit
Tc	Operating temperature	-40	-	85	°C

7.4 Peripheral interfaces

7.4.1 SPI master interface characteristics

- The operating characteristics of the SPI master controller at typical temperature (25°C) are shown in Figure 7.4-1, Figure 7.4-2 and Table 7.4-1. CPOL defines the clock polarity in the transmission. CPHA defines the legal timing to sample data.

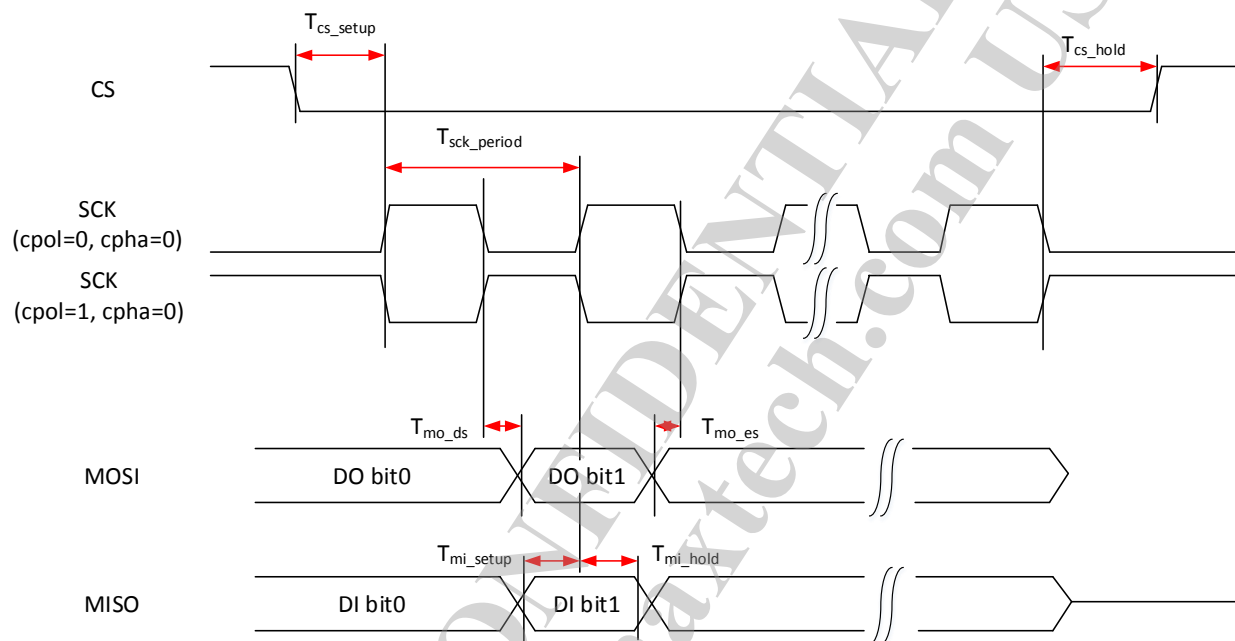


Figure 7.4-1. SPI master interface timing diagram (CPHA=0)

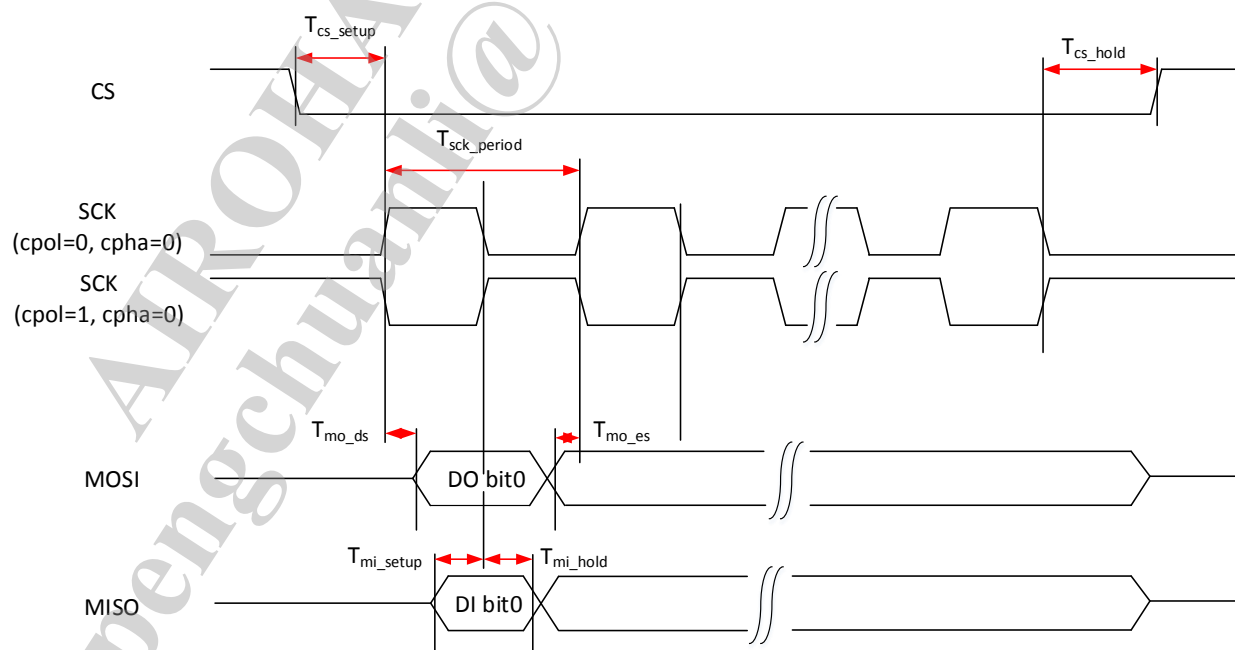


Figure 7.4-2. SPI master interface timing diagram (CPHA=1)

Table 7.4-1. SPI master interface characteristics

Signal	Symbol	Parameter	Min.	Max.	Unit	Description
CS	T_{cs_setup}	Chip select setup time	9.6	-	ns	
	T_{cs_hold}	Chip select hold time	28.8	-	ns	
SCK	T_{sck_period}	Serial clock period	19.2	-	ns	
MOSI	T_{mo_ds}	SPI master data output delay skew	-	1.0	ns	
	T_{mo_es}	SPI master data output early skew	-	1.0	ns	
MISO	T_{mi_setup}	SPI master data input valid time	4.8	-	ns	
	T_{mi_hold}	SPI master data input hold time	4.8	-	ns	

7.4.2 SPI slave interface characteristics

- The operating characteristics of the SPI slave at typical temperature (25°C) are shown in Figure 7.4-3, Figure 7.4-4 and Table 7.4-2. CPOL defines the clock polarity in the transmission. CPHA defines the legal timing to sample data.

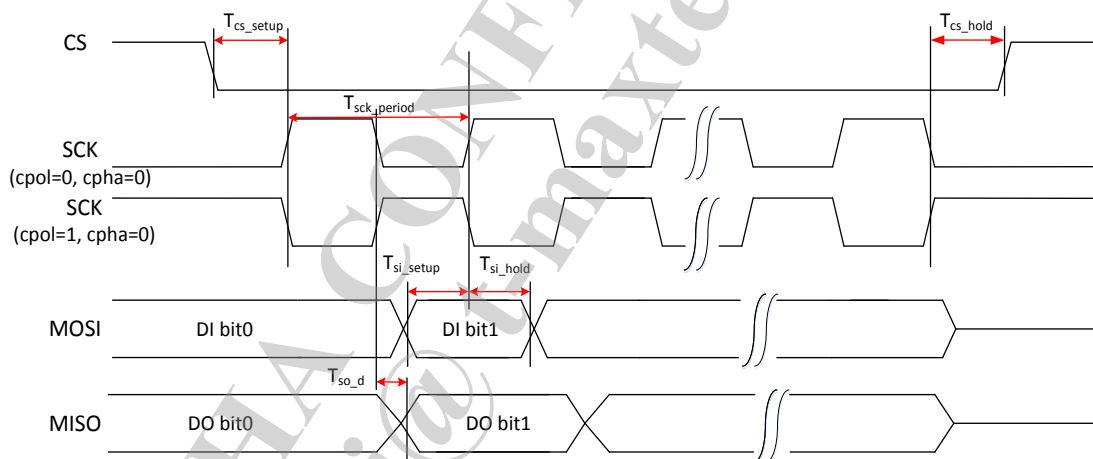


Figure 7.4-3. SPI slave interface timing diagram (CPHA=0)

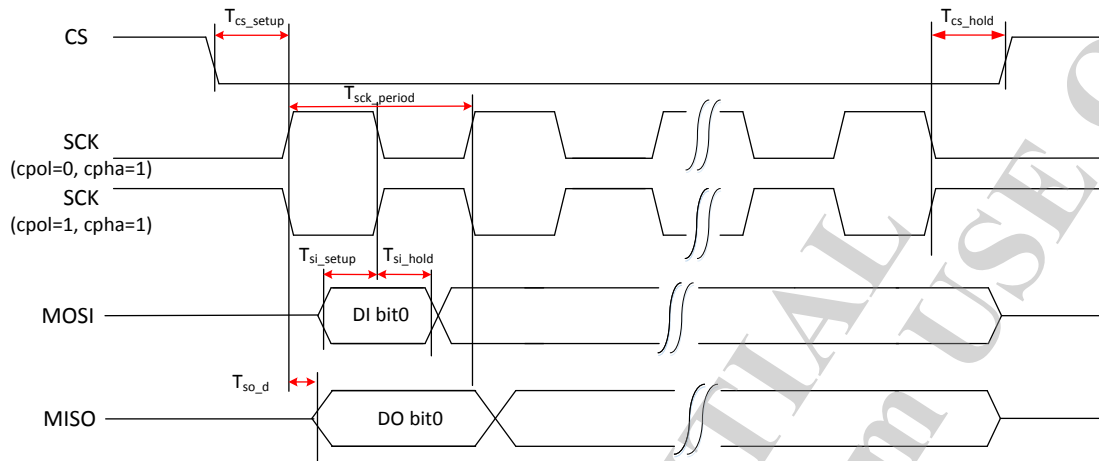


Figure 7.4-4. SPI slave interface timing diagram (CPHA=1)

Table 7.4-2. SPI slave interface characteristics

Signal	Symbol	Parameter	Min.	Max.	Unit	Description
CS	T_{cs_setup}	Chip select setup time	9.6	-	ns	
	T_{cs_hold}	Chip select hold time	28.8	-	ns	
SCK	T_{sck_period}	Serial clock period	19.2	-	ns	
MOSI	T_{si_setup}	SPI slave data input valid time	4.8	-	ns	
	T_{si_hold}	SPI slave data input hold time	4.8	-	ns	
MISO	T_{so_d}	SPI slave data output delay	-	15	ns	

7.4.3 SDIO2.0 master interface characteristics

7.4.3.1 DS mode

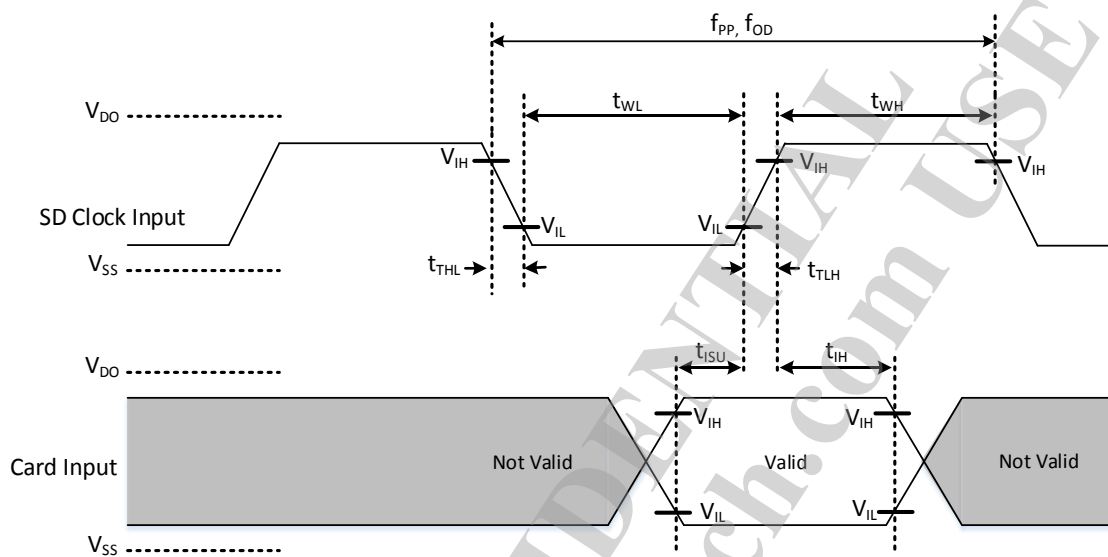


Figure 7.4-5. Card Input Timing (Default Speed Mode)

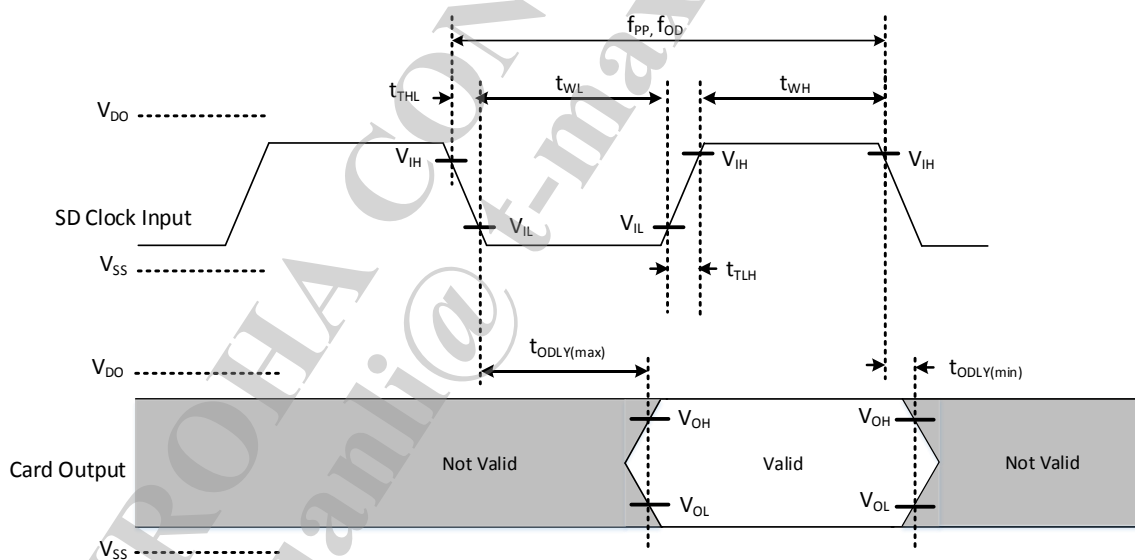


Figure 7.4-6. Card Output Timing (Default Speed Mode)

Table 7.4-3. Bus Timing – Parameters Values (Default Speed)

Parameter	Symbol	Min.	Max.	Unit	Remark
Clock CLK(All values are referred to min(VIH) and max(VIL))					
Clock frequency Data Transfer Mode	fPP	0	24	MHz	CCARD ≤ 10 pF (1 card)

Parameter	Symbol	Min.	Max.	Unit	Remark
Clock frequency Identification Mode	fOD	0 ⁽¹⁾ /100	400	kHz	CCARD ≤ 10 pF (1 card)
Clock low time	tWL	10	–	ns	CCARD ≤ 10 pF (1 card)
Clock high time	tWH	10	–	ns	CCARD ≤ 10 pF (1 card)
Clock rise time	tTLH	–	10	ns	CCARD ≤ 10 pF (1 card)
Clock fall time	tTHL	–	10	ns	CCARD ≤ 10 pF (1 card)
Input CMD,DAT (referenced to CLK)					
Input setup time	tISU	5	–	ns	CCARD ≤ 10 pF (1 card)
Input hold time	tIH	5	–	ns	CCARD ≤ 10 pF (1 card)
Output CMD,DAT (referenced to CLK)					
Output Delay time during Data Transfer Mode	tODLY	0	14	ns	CL ≤ 40 pF (1 card)
Output Delay time during Identification Mode	tODLY	0	50	ns	CL ≤ 40 pF (1 card)

(1) 0 Hz means to stop the clock. The given minimum frequency range is for cases where continuous clock is required.

7.4.3.2 HS mode

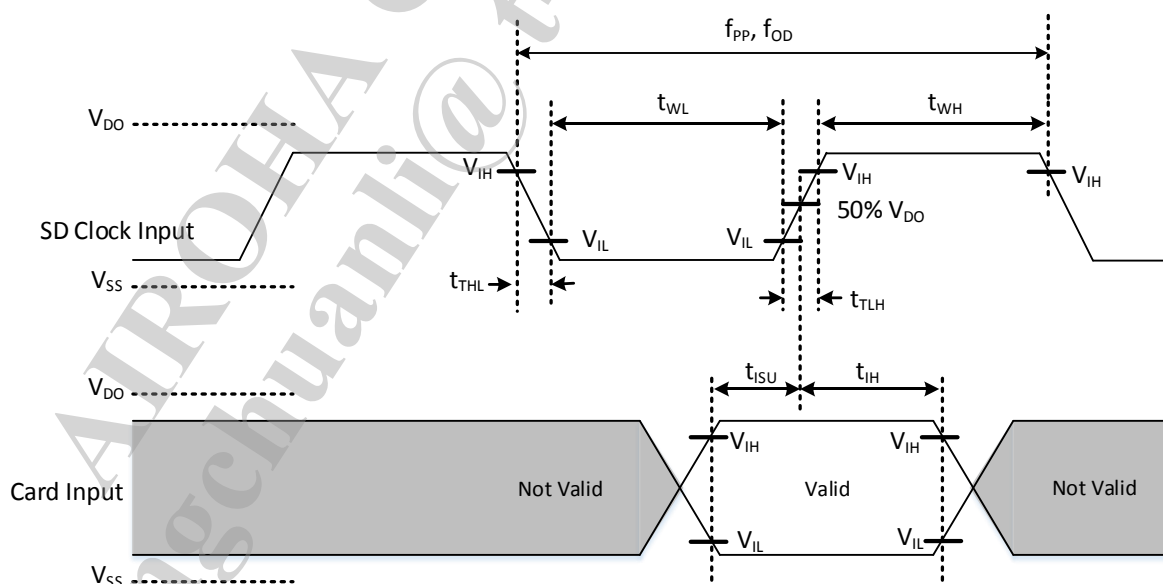


Figure 7.4-7. Card Input Timing (High Speed Mode)

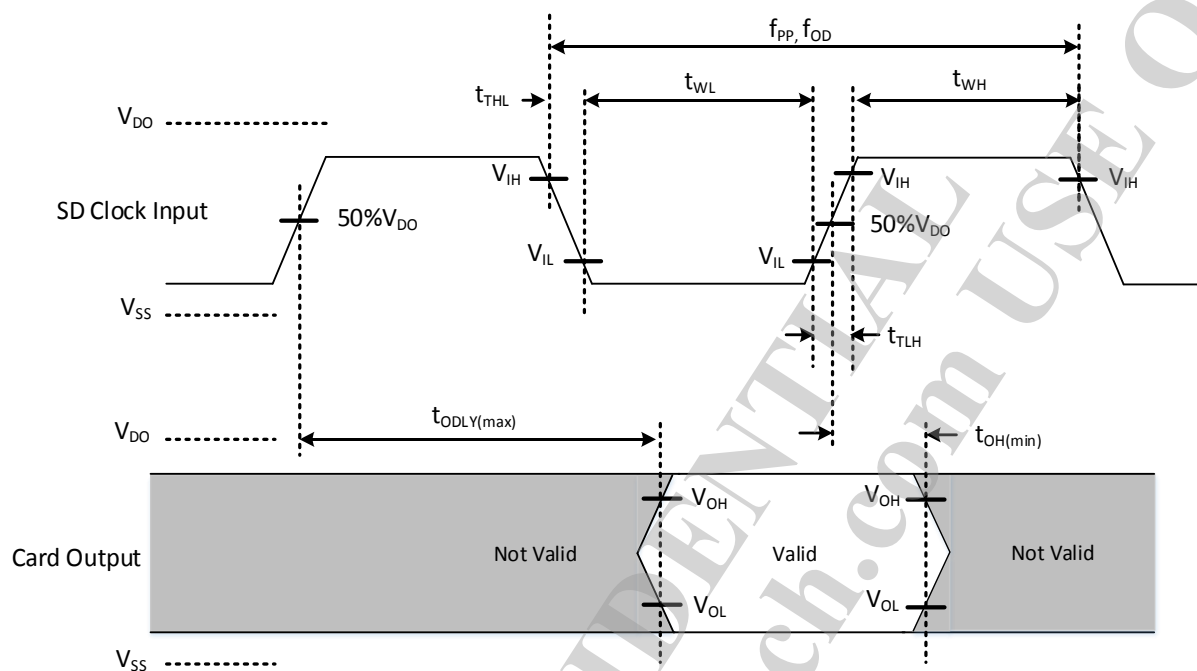


Figure 7.4-8. Card Output Timing (High Speed Mode)

Table 7.4-4. Bus Timing – Parameters Values (High Speed)

Parameter	Symbol	Min.	Max.	Unit	Remark
Clock CLK(All values are referred to min(VIH) and max(VIL))					
Clock frequency Data Transfer Mode	fPP	0	48	MHz	CCARD ≤ 10 pF (1 card)
Clock frequency Identification Mode	fOD	0 ⁽¹⁾ /100	400	kHz	CCARD ≤ 10 pF (1 card)
Clock low time	tWL	7	–	ns	CCARD ≤ 10 pF (1 card)
Clock high time	tWH	7	–	ns	CCARD ≤ 10 pF (1 card)
Clock rise time	tTLH	–	3	ns	CCARD ≤ 10 pF (1 card)
Clock fall time	tTHL	–	3	ns	CCARD ≤ 10 pF (1 card)
Input CMD,DAT (referenced to CLK)					
Input setup time	tISU	6	–	ns	CCARD ≤ 10 pF (1 card)
Input hold time	tIH	2	–	ns	CCARD ≤ 10 pF (1 card)
Output CMD,DAT (referenced to CLK)					

Parameter	Symbol	Min.	Max.	Unit	Remark
Output Delay time during Data Transfer Mode	tODLY	–	14	ns	CL ≤ 40 pF (1 card)
Output Hold time	tOH	2.5	–	ns	CL ≥ 15 pF (1 card)

(1) 0 Hz means to stop the clock. The given minimum frequency range is for cases where continuous clock is required.

7.4.4 eMMC interface characteristics

- The I/O operating voltage characteristics of the eMMC controller at typical temperature (25°C) is shown in Table 7.4-5 and Table 7.4-6.

Table 7.4-5. eMMC interface push-pull signal level----2.7V – 3.6V V_{CCQ} range (high voltage eMMC)

Parameter	Symbol	Minimum	Maximum	Unit	Description
Output High Voltage	V _{OH}	0.75*V _{CCQ}	–	V	I _{OH} = -100μA at V _{CCQ} min
Output Low Voltage	V _{OL}	–	0.125*V _{CCQ}	V	I _{OL} = 100μA at V _{CCQ} min
Input High Voltage	V _{IH}	0.625*V _{CCQ}	V _{CCQ} +0.3	V	
Input Low Voltage	V _{IL}	V _{SS} -0.3	0.25*V _{CCQ}	V	

Table 7.4-6. eMMC interface push-pull signal level----1.70V – 1.95V V_{CCQ} voltage range

Parameter	Symbol	Minimum	Maximum	Unit	Description
Output High Voltage	V _{OH}	V _{CCQ} -0.45V	–	V	I _{OH} = -2mA
Output Low Voltage	V _{OL}	–	0.45V	V	I _{OL} = 2mA
Input High Voltage	V _{IH}	0.65*V _{CCQ} ⁽¹⁾	V _{CCQ} +0.3	V	
Input Low Voltage	V _{IL}	V _{SS} -0.3	0.35*V _{CCQ} ⁽²⁾	V	

- ⁽¹⁾0.7*V_{DD} for MMC4.3 and older revisions
- ⁽²⁾0.3*V_{DD} for MMC4.3 and older revisions
- The eMMC bus timing is shown in Figure 7.4-9, Table 7.4-7 and Table 7.4-8.

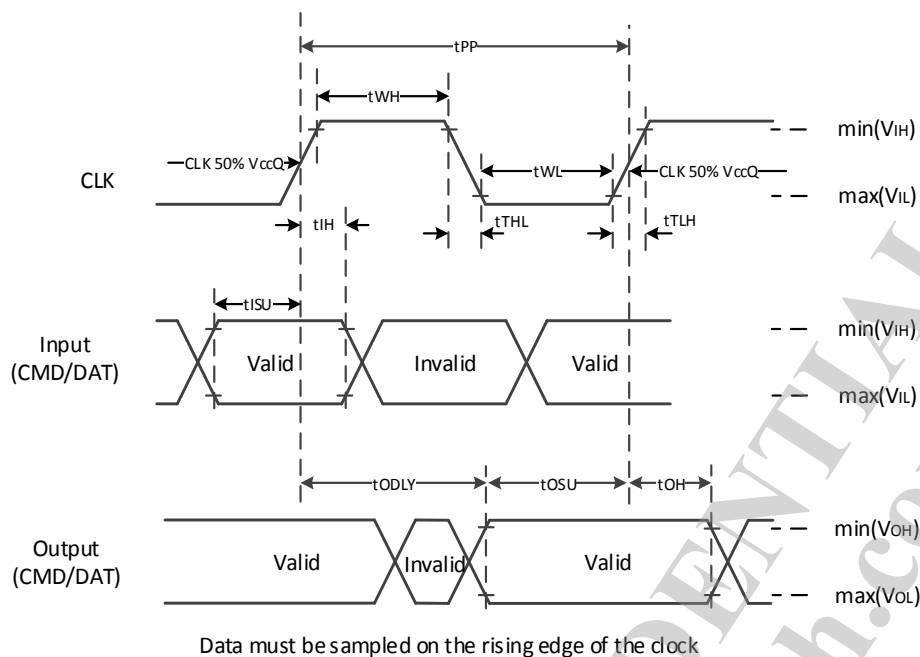


Figure 7.4-9. eMMC interface timing diagram

Table 7.4-7. eMMC interface characteristics (backward-compatible device)

Parameter	Symbol	Minimum	Maximum	Unit	Description ⁽¹⁾
Clock CLK ⁽²⁾					
Clock frequency data transfer Mode (PP) ⁽³⁾	f_{pp}	0	26	MHz	$CL \leq 30pF$
Clock frequency Identification Mode (OD)	f_{od}	0	400	kHz	
Clock low time	t_{WL}	10	—	ns	$CL \leq 30pF$
Clock high time	t_{WH}	10	—	ns	$CL \leq 30pF$
Clock rise time ⁽⁴⁾	t_{TLH}	—	10	ns	$CL \leq 30pF$
Clock fall time	t_{THL}	—	10	ns	$CL \leq 30pF$
Inputs CMD, DAT (referenced to CLK)					
Input set-up time	t_{ISU}	3	—	ns	$CL \leq 30pF$
Input hold time	t_{IH}	3	—	ns	$CL \leq 30pF$
Outputs CMD, DAT (referenced to CLK)					
Output set-up time ⁽⁵⁾	t_{OSU}	11.7	—	ns	$CL \leq 30pF$
Output hold time ⁽⁵⁾	t_{OH}	8.3	—	ns	$CL \leq 30pF$

- ⁽¹⁾The device must always start with the backward-compatible interface timing. The timing mode can be switched to high-speed interface timing by the host sending the SWITCH command (CMD6) with the argument for high-speed interface select.
- ⁽²⁾CLK timing is measured at 50% of V_{CCQ}

- ⁽³⁾For compatibility with devices that support the v4.2 standard or earlier, host should not use frequency greater than 26MHz before switching to high-speed interface timing.
- ⁽⁴⁾CLK rise and fall times are measured by min(V_{IH}) and max(V_{IL})
- ⁽⁵⁾t_{OSU} and t_{OH} are defined as values from clock rising edge. However, there may be devices which utilize clock falling edge to output data in backward compatibility mode. Therefore, it is recommended for hosts either to set t_{WL} values as long as possible within the range which will not go over t_{CK-tOH(min)} in the system or to use slow clock frequency, so that host could have data set up margin for those devices. In this case, each device which utilizes clock falling edge might show the correlation either between t_{WL} and t_{OSU} or between t_{CK} and t_{OSU} for the device in its own datasheet as a note or an application notes.

Table 7.4-8. eMMC interface characteristics (high-speed device)

Parameter	Symbol	Min.	Max.	Unit	Description
Clock CLK⁽¹⁾					
Clock frequency Data Transfer Mode (PP) ⁽²⁾	f _{pp}	0	52 ⁽³⁾	MHz	C _L ≤ 30pF Tolerance: +100kHz
Clock frequency Identification Mode (OD)	f _{OD}	0	400	kHz	Tolerance: +20kHz
Clock low time	t _{WL}	6.5	—	ns	C _L ≤ 30pF
Clock high time	t _{WH}	6.5	—	ns	C _L ≤ 30pF
Clock rise time ⁽⁴⁾	t _{TLH}	—	3	ns	C _L ≤ 30pF
Clock fall time	t _{THL}	—	3	ns	C _L ≤ 30pF
Inputs CMD, DAT (referenced to CLK)					
Input set-up time	t _{ISU}	3	—	ns	C _L ≤ 30pF
Input hold time	t _{IH}	3	—	ns	C _L ≤ 30pF
Outputs CMD, DAT (referenced to CLK)					
Output delay time during data transfer	t _{ODLY}	—	13.7	ns	C _L ≤ 30pF
Output hold time	t _{OH}	2.5	—	ns	C _L ≤ 30pF
Signal rise time ⁽⁵⁾	T _{RISE}	—	3	ns	C _L ≤ 30pF
Signal fall time	T _{FALL}	—	3	ns	C _L ≤ 30pF

- ⁽¹⁾CLK timing is measured at 50% of V_{CCQ}
- ⁽²⁾An eMMC will support the full frequency range from 0MHz – 26MHz, or 0MHz – 52MHz
- ⁽³⁾Devices can operate as high-speed device interface timing at 26MHz clock frequency.
- ⁽⁴⁾CLK rise and fall times are measured by min(V_{IH}) and max(V_{IL}).
- ⁽⁵⁾Inputs CMD, DAT rise and fall times are measured by min(V_{IH}) and max(V_{IL}), and outputs CMD, DAT rise and fall times are measured by min(V_{OH}) and max(V_{OL}).

7.4.5 USB2.0 interface characteristics

7.4.5.1 FS/LS mode

- The USB uses a differential output driver to drive the USB data signal on the USB cable. The static output swing of the driver in its low state must be below VOL (max) of 0.3V with a 1.5kΩ load to 3.6V and in its high state must be above the VOH (min) of 2.8V with a 15kΩ load to ground.

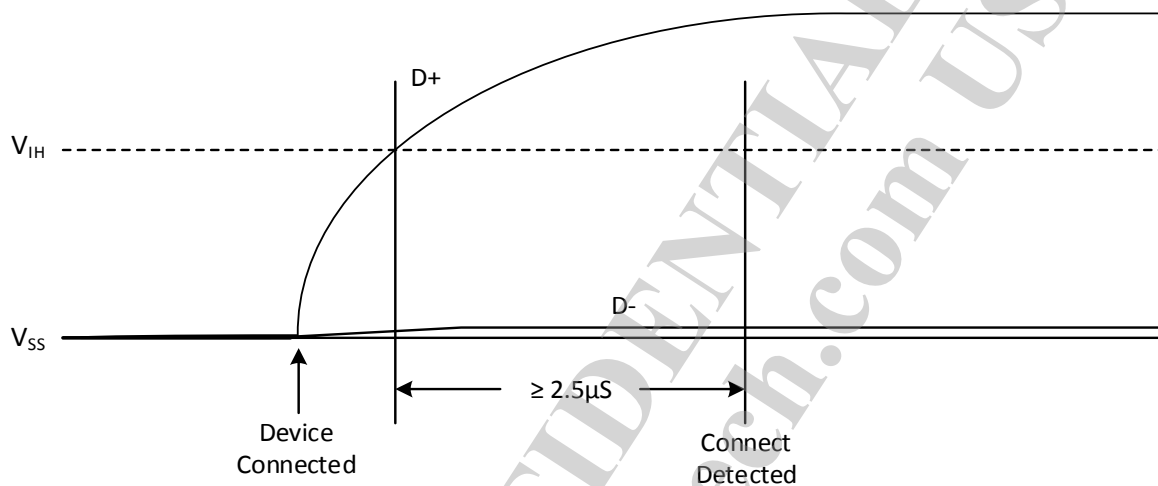


Figure 7.4-10. Full-speed Device Connect Detection

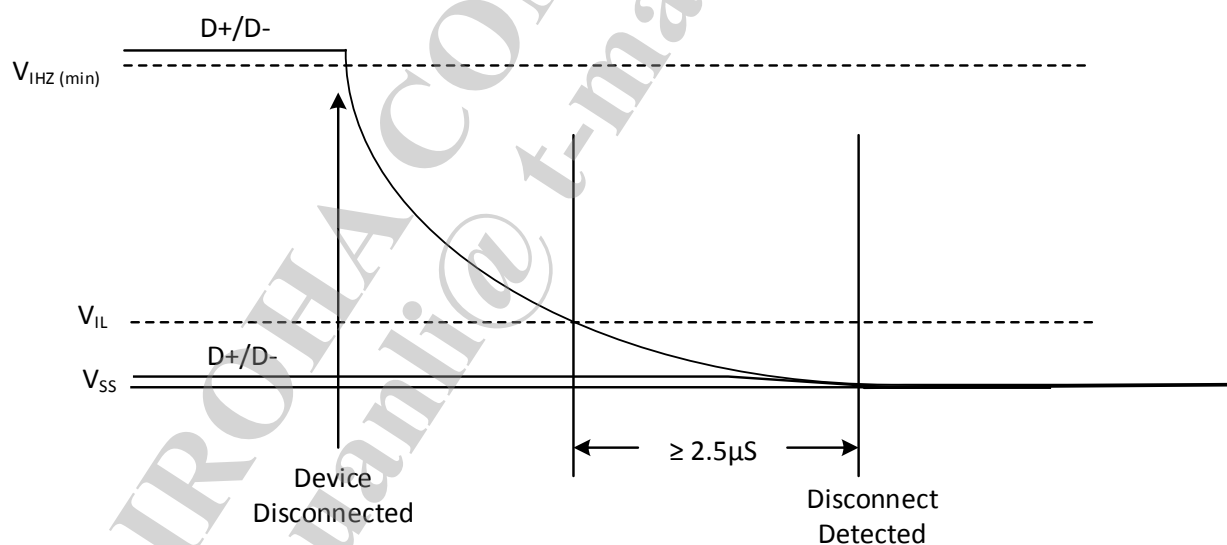


Figure 7.4-11. Disconnect Detection

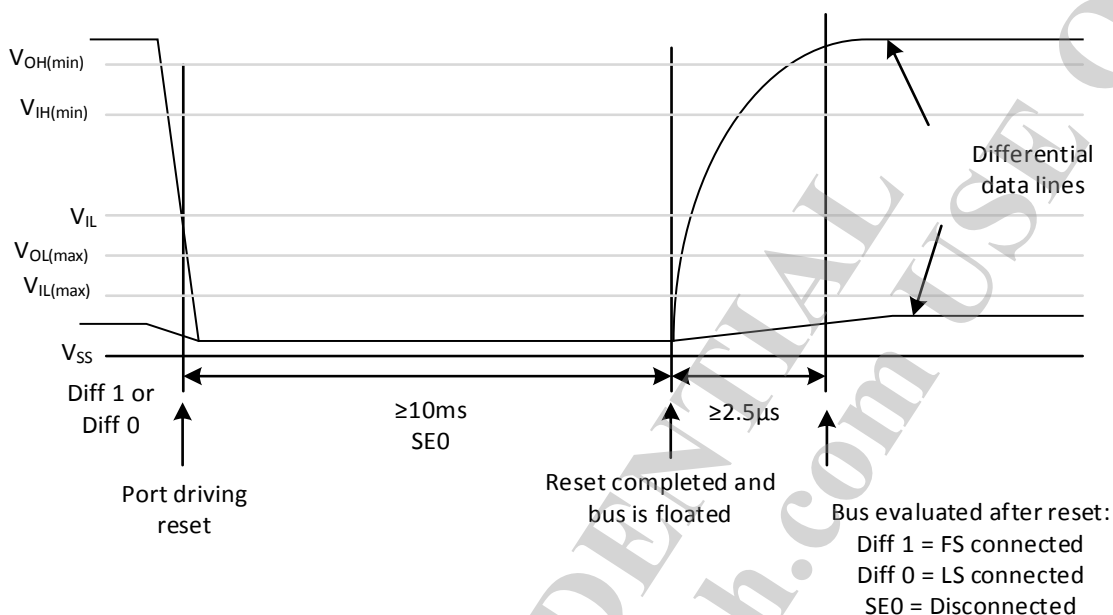


Figure 7.4-12. Bus State Evaluation after reset

7.4.5.2 HS mode

- High-speed capable devices must initially attach as full-speed devices and must comply with all full-speed connection requirements. Transition to high-speed signaling is accomplished by means of a low level electrical protocol which occurs during Reset. The differential output impedance of a high-speed capable driver is required to be $90\Omega \pm 10\%$. When either the D+ or D- lines are driven high, VHSOH (the high-speed mode high-level output voltage driven on a data line with a precision 45Ω load to GND) must be $400\text{ mV} \pm 10\%$. On a line which is not driven, either because the transceiver is not transmitting or because the opposite line is being driven high, VHSOL (the high-speed mode low-level output voltage driven on a data line with a 45Ω load to GND) must be $0\text{ V} \pm 10\text{ mV}$.

Table 7.4-9. USB20 interface characteristics

Signal	Symbol	Parameter	Min.	Max.	Unit	Description
D+/D-	VIH	High (driven)	2	-	V	
	VIHZ	High (floating)	2.7	3.6	V	
	VIL	Low	-	0.8	V	
	VOH	High (driven)	2.8	3.6	V	
	VOL	Low	0.0	0.3	V	
	VHSOH	High-speed data signaling high	360	440	mV	
	VHSOL	High-speed data signaling low	-10	10	mV	

7.4.6 External serial flash interface characteristics

- The operating characteristics of the SF controller at typical temperature (25°C) are shown in Figure 7.4-13.

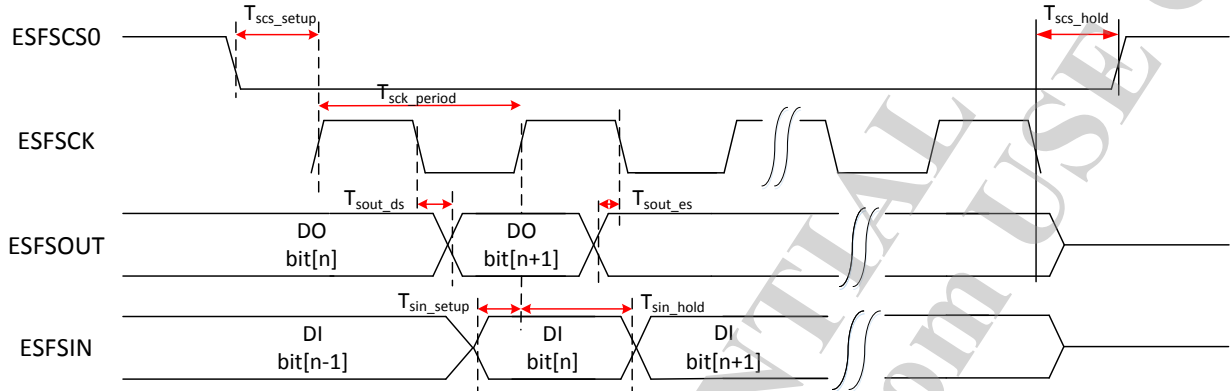


Figure 7.4-13. SF controller interface timing

Signal	Symbol	Parameter	Min.	Max.	Unit	Description
ESFCS0	T_{scs_setup}	Chip select setup time	10	-	ns	
	T_{scs_hold}	Chip select hold time	10	-	ns	
ESFSCK	T_{sck_period}	Serial clock period	13	-	ns	
ESFSOUT	T_{sout_ds}	SFC data output delay skew	-	3	ns	
	T_{sout_es}	SFC data output early skew	-	3	ns	
ESFSIN	T_{sin_setup}	SFC data input valid time	3	-	ns	
	T_{sin_hold}	SFC data input hold time	3	-	ns	

7.4.7 External SRAM Controller Interface characteristics

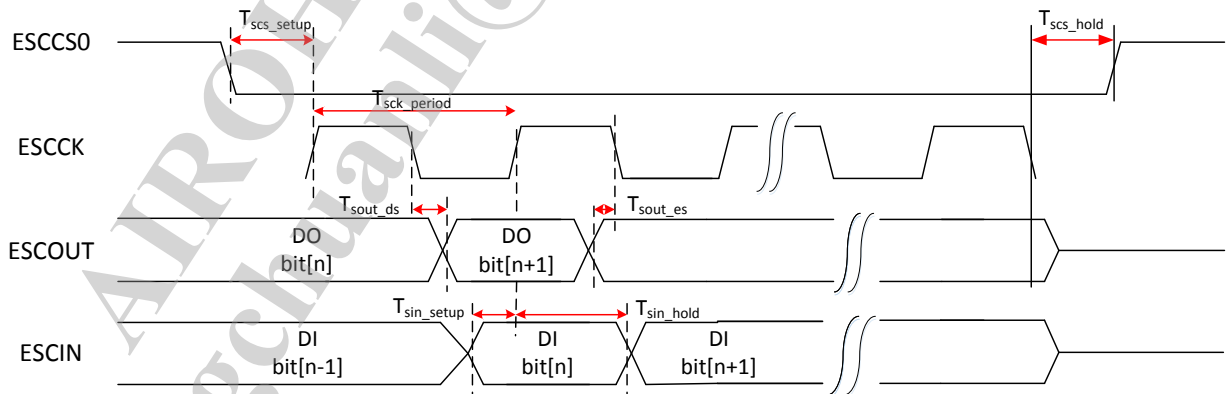
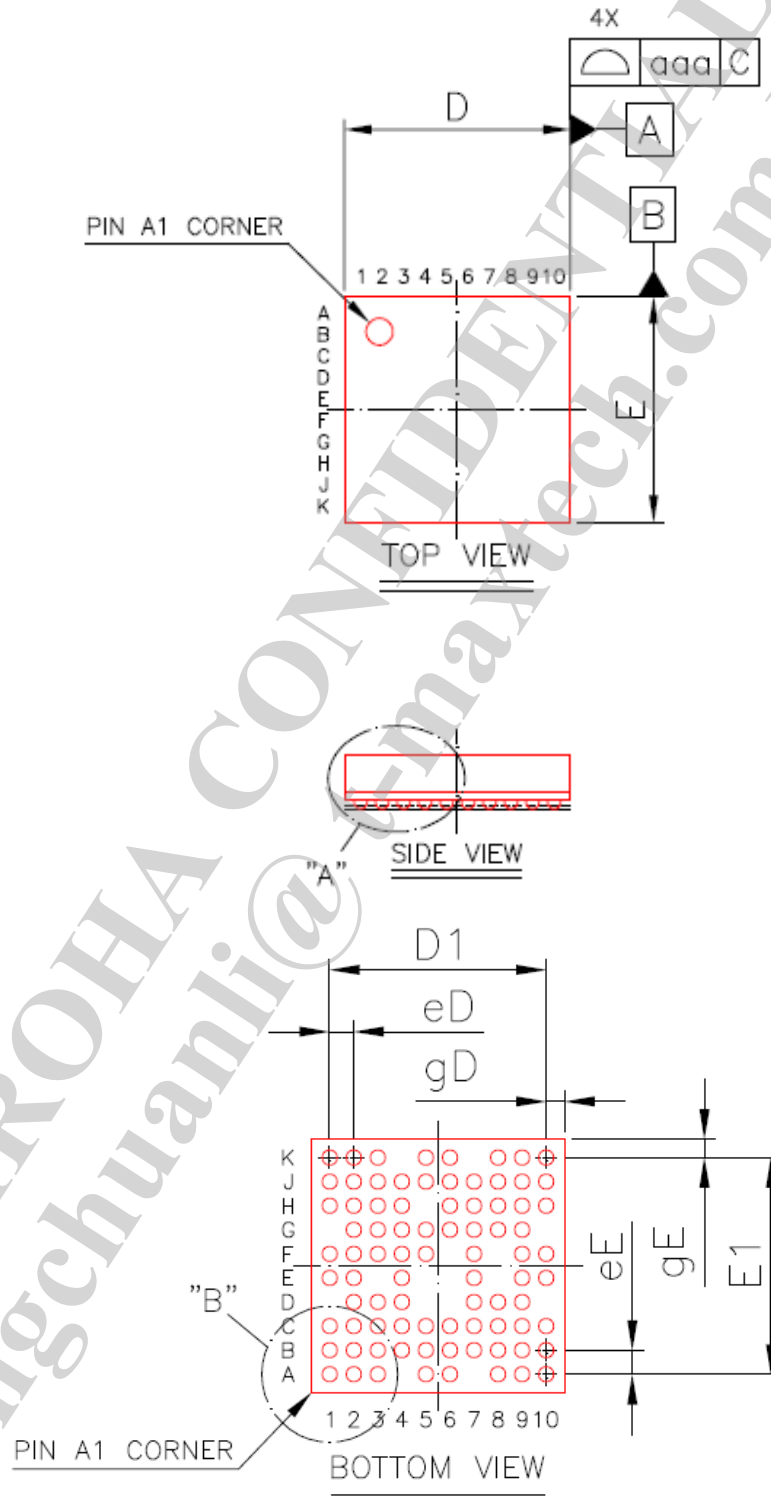


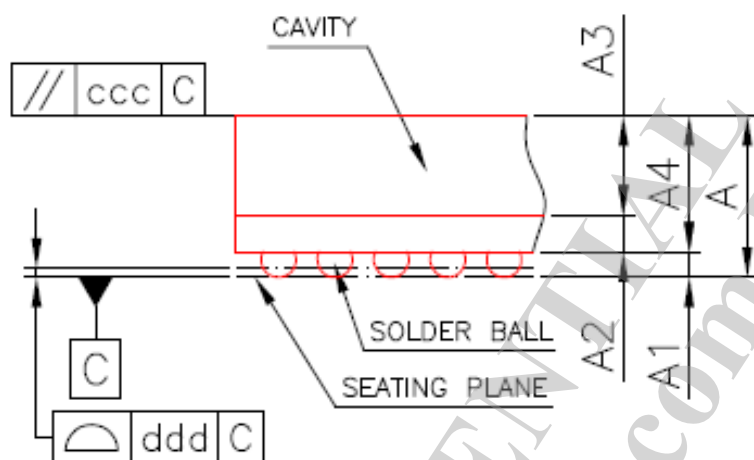
Figure 7.4-14 ESC interface timing

Signal	Symbol	Parameter	Min.	Max.	Unit	Description
ESCCS0	T _{scs_setup}	Chip select setup time	10	-	ns	
	T _{scs_hold}	Chip select hold time	10	-	ns	
ESCCK	T _{sck_period}	Serial clock period	38.47	-	ns	
ESCOUT	T _{sout_ds}	ESC data output delay skew	-	10	ns	
	T _{sout_es}	ESC data output early skew	-	10	ns	
ESCIN	T _{sin_setup}	ESC data input valid time	10	-	ns	
	T _{sin_hold}	ESC data input hold time	10	-	ns	

8 Package Information

8.1 AG3335A mechanical data of the package

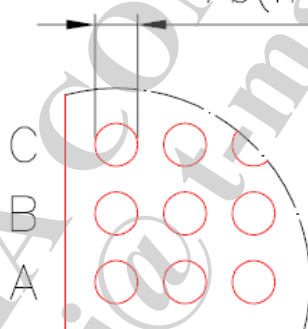




DETAIL : "A"

Φ	$\varnothing e e e (M)$	C	A	B
	$\varnothing f f f (M)$	C		

$\varnothing b(n \times)$



DETAIL : "B"

Item		Symbol	Common Dimensions		
			MIN.	NOM.	MAX.
Package Type			MFW TFBGA		
Body Size	X	D	4.150	4.200	4.250
	Y	E	4.150	4.200	4.250
Ball Pitch	X	eD	0.400		
	Y	eE	0.400		
Mold Thickness		A3	0.700 Ref.		
Substrate Thickness		A2	0.141 Ref.		
Substrate+Mold Thickness		A4	0.791	0.841	0.891
Total Thickness		A	—	—	1.066
Ball Diameter			0.250		
Ball Stand Off		A1	0.120	0.160	0.200
Ball Width		b	0.200	0.250	0.300
Package Edge Tolerance		aaa	0.050		
Mold Flatness		ccc	0.100		
Coplanarity		ddd	0.080		
Ball Offset (Package)		eee	0.150		
Ball Offset (Ball)		fff	0.050		
Ball Count		n	83		
Edge Ball Center to Center	X	D1	3.600		
	Y	E1	3.600		
Edge Ball Center to Package Edge	X	gD	0.300		
	Y	gE	0.300		

Figure 8.1-1. Outlines and dimensions of AG3335A TFBGA 4.2 mm*4.2 mm, 83-ball, 0.4mm pitch package

8.2 AG3335A thermal operating specifications

Table 8.2-1. AG3335A thermal characteristics without external heat sink in still air condition

Description	Value	Unit
Thermal Resistance θ_{JA} ($^{\circ}\text{C}/\text{W}$) for JEDEC 4L system PCB	42.56	$^{\circ}\text{C}/\text{W}$
Thermal Resistance θ_{JC} ($^{\circ}\text{C}/\text{W}$) for JEDEC system PCB	15.15	$^{\circ}\text{C}/\text{W}$
Maximum power dissipation	0.998	W

NOTE: JEDEC 51-9 system FR4 PCB size: 101.5x114.5mm (4"x4.5")

9 Ordering Information

9.1 AG3335A top marking

The product part number is AG3335A.



Line1: Airoha logo

Line2: ARM logo

Line3: device name

Line4: date code & "\$" assembly code

Line5: U1 wafer lot

Line6: U2 & U3 wafer

Table 9.1-1. Ordering information

Product number	Package	Description
AG3335AA/B	TFBGA	4.2mm*4.2mm, 83-ball, 0.4mm pitch Package